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Realization of a Data Center Modern Network Architecture

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CONTENTS

LIST OF ACRONYMS

Acronym	Meaning
CPU	Central Processing Unit
DMZ	Demilitarized Zone
EoR	End of Row
HLD	High-Level Design
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IP	Internet Protocol
IT	Information Technology
LLD	Low-Level Design
MLAG	Multi-Chassis Link Aggregation
NAT	Network Address Translation
NIC	Network Interface Card
OS	Operating System
SSH	Secure Shell
ToR	Top of Rack
UML	Unified Modeling Language
VLAN	Virtual Local Area Network
VM	Virtual Machine
VRRP	Virtual Router Redundancy Protocol

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GENERAL INTRODUCTION

In an era defined by the exponential growth of digital data and the increasing demand for highly available computing infrastructure, data centers have become the cornerstone of modern information systems. Their design is no longer a purely physical engineering task but a complex discipline combining network architecture, virtualization, and security.

Modern enterprises increasingly require infrastructure that is highly available, scalable, secure, and centrally manageable to reliably deliver digital services at scale. Traditional three-tier data center architectures suffer from limitations such as high east-west latency, limited scalability, and potential single points of failure. To address these challenges, this work adopts a Spine-Leaf architecture combined with a layered security model separating Trust and DMZ zones through redundant firewalls. The project also implements multi-level redundancy and validates network resilience through failure testing, resulting in a fully simulated and production-oriented data center design.

This dissertation, prepared for the Licence degree in ISIL at USTHB, addresses the design and simulation of a modern data center architecture. Rather than working with physical equipment, this project constructs a complete virtual environment using PNETLab to emulate network devices (core switches, firewalls, and ToR switches) and VMware to virtualize servers across four cabinets.

The adopted architecture is a Spine-Leaf model, combining the full-mesh connectivity of Spine-Leaf with the security structure of the traditional three-tier model. Four cabinets (A, B, C, and D) contain eight ToR leaf switches interconnected through two EoR spine switches. Two firewalls enforce strict zone separation between the Trust zone (Cabinets A, B, and C) and the DMZ (Cabinet D).

This dissertation is organized into three chapters:

Chapter 1: State of the Art presents fundamental concepts of data centers, their evolution, key components, and architectural models including three-tier, Spine-Leaf, and the model adopted.

Chapter 2: Conception and Design details the system design, including actor identification, functional and non-functional requirements, UML diagrams (use case and sequence), VLAN segmentation, IP addressing, VRRP, MLAG, security zones, and HLD/LLD documentation.

Chapter 3: Implementation and Simulation covers the practical realization, including device configuration in PNETLab, VMware virtualization setup, connectivity tests, and redundancy validation.

We conclude by summarizing the work accomplished and outlining perspectives for future improvement.

CHAPTER 1 STATE OF THE ART

1.1 Introduction

This chapter presents the fundamental concepts necessary to understand our data center design. We first define what a data center is and its key components. Then we compare traditional Three-Tier architecture with modern Spine-Leaf architecture. Finally, we introduce the core protocols and tools used in this project.

1.2 Overview of Data Centers

1.2.1 Definition and Role

A data center is a physical facility that organizations use to house their critical applications and data. A data center's design is based on a network of computing and storage resources that enable the delivery of shared applications and data. The key components of data center design include routers, switches,firewalls, servers, and application-delivery controllers [1]. Its four essential components are:

- **Compute:** Physical or virtual servers that process applications and services. In modern data centers, servers are increasingly virtualized using hypervisors such as VMware.
- **Storage:** Systems responsible for storing data, including SAN (Storage Area Network), NAS (Network Attached Storage), and distributed storage solutions.
- **Network:** The backbone of the data center, composed of switches, routers, and firewalls that interconnect all components and control traffic flow between them.
- **Power and cooling:** Uninterruptible power supplies, generators, and cooling systems that ensure the physical environment remains operational at all times [2].

1.2.2 Huawei: From Telecom Equipment Manufacturer to Data Center Leader

Huawei is a Chinese multinational technology company founded in 1987, originally focused on telecommunications equipment before expanding into enterprise networking and cloud infrastructure. Today, Huawei is a leading provider of end-to-end data center solutions, offering switching fabrics, storage systems, and cloud platforms deployed in carrier-grade facilities worldwide [3].

1.3 Data Center Architectures

1.3.1 Data Center Network Architecture

1.3.2 Three-Tier Architecture

The three-tier network architecture consists of three layers: the **core layer**, which provides high-speed backbone connectivity; the **aggregation layer**, which manages traffic and network policies; and the **access layer**, where servers and storage devices connect to the network. Together, these layers improve network performance, reliability, and scalability [4].

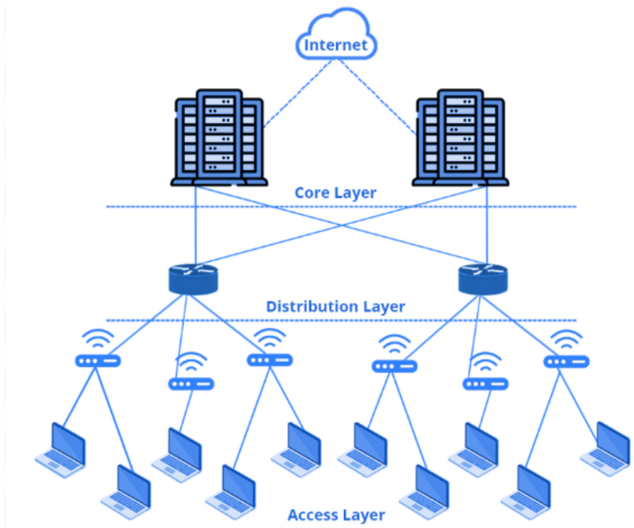


Figure 1.1: *3-Tier Architecture*

1.3.3 Spine-Leaf Architecture

The Spine-Leaf architecture is a modern two-tier model designed to overcome the limitations of the three-tier approach. It consists of spine switches that form the backbone and leaf switches that connect to end devices, with every leaf switch connected to every spine switch in a full mesh. This design guarantees that any two devices in the network are always separated by exactly two hops, delivering consistent low latency, high bandwidth, and horizontal scalability. It is widely adopted in hyperscale and cloud data centers [5].

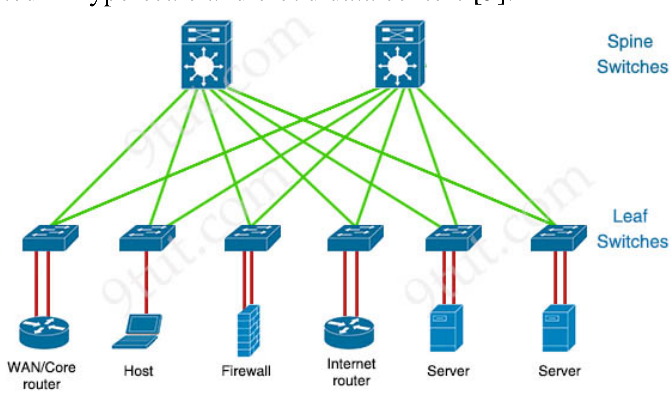


Figure 1.2: *Spine-Leaf Architecture*

1.4 Network Components

1.4.1 Spine Switches

Spine switches form the core forwarding layer in a spine-leaf architecture. Therefore, EoR-1 and EoR-2 serve as spine switches. They connect only to leaf switches and do not connect directly to servers. Their primary role is to provide high-speed, redundant connectivity between the leaf switches, ensuring reliable and scalable traffic forwarding across the fabric.

Figure 1.3: *Huawei CloudEngine S5731 Switch Model*

1.4.2 Leaf Switches (ToR):

Leaf switches provide server connectivity in the access layer. Positioned as Top-of-Rack switches, they connect to servers via access ports and uplink to spine switches via trunk ports. Each cabinet typically contains two leaf switches operating in active-standby mode using MLAG, with one ToR active and one standby, providing sub-second failover [6].

1.4.3 Firewalls:

Firewalls enforce security policies between network zones. In our design, two firewalls (FW-1 and FW-2) operate in an active-standby cluster at the network edge. They provide zone separation between Untrust (external networks), DMZ, and Trust zones, as well as stateful packet inspection tracking connection states. Firewalls are critical for enforcing the principle of least privilege: the DMZ can receive external traffic, but the Trust zone can only be accessed through controlled firewall rules [7].

Figure 1.4: *Huawei HiSecEngine USG6700E Firewall Model*

1.4.4 Servers:

Servers host the applications and data that a data center exists to serve. The separation of management and service networks follows security best practices: administrators access management IPs for configuration, while service IPs handle application traffic. This isolation prevents management access from being exposed to the application layer [8].

Figure 1.5: *Dell PowerEdge R760xa Server Model*

1.5 Core Networking Concepts

1.5.1 VLAN Segmentation and Network Zones

A VLAN logically segments a physical network into isolated broadcast domains. This improves security and reduces unnecessary traffic. In data centers, VLANs separate different traffic types such as management, storage, and application traffic. Network zones (DMZ, Trust, Untrust) extend VLAN isolation with firewall-enforced security policies. The DMZ hosts public-facing services, the Trust zone contains internal applications, and the Untrust zone represents external networks [9].

1.5.2 Redundancy and High Availability

Redundancy is a fundamental principle in data center design. It ensures that the failure of a single component does not result in a service outage. In our project, redundancy is implemented at every layer of the architecture [10]:

- **Security layer:** Two firewalls deployed in parallel, both connected to external networks and to the core switches.
- **Spine layer:** Two EoR switches, each connected to all eight ToR switches, providing multiple forwarding paths.
- **Leaf layer:** Each cabinet contains two ToR switches, ensuring servers remain connected even if one switch fails.

This multi-level redundancy guarantees high availability across the entire infrastructure using multiple protocols.

1.5.3 Virtualization Principles

Virtualization is the process of creating virtual instances of physical resources such as servers and networks. VMware is used to virtualize the IT infrastructure within each cabinet, creating virtual servers that communicate with the simulated network in PNETLab [11].

1.6 Conclusion

This chapter has presented the essential concepts of modern data centers, comparing traditional Three-Tier architecture with Spine-Leaf. The analysis shows that Spine-Leaf offers consistent latency, better east-west traffic handling, and horizontal scalability, making it the preferred choice for cloud-scale infrastructures. These findings provide the theoretical basis for the design presented in the next chapter.

CHAPTER 2 CONCEPTION AND DESIGN

2.1 Introduction

This chapter translates the theoretical spine-leaf architecture from Chapter 1 into a concrete design. We detail the topology, VLANs, IP addressing, VRRP, MLAG, security zones, and UML models. The resulting blueprint guides implementation in Chapter 3.

2.2 Architecture Overview

2.2.1 Spine-Leaf Topology (HLD)

The architecture adopts a Spine-Leaf model across four virtual cabinets: A, B, C, and D. Each cabinet is equipped with two ToR switches acting as leaf nodes, giving a total of eight ToR switches at the access layer. The cabinets are interconnected through two EoR core switches acting as spine nodes, each connected to all eight ToR switches, ensuring that any server can reach any other server with minimal hops and maximum redundancy.

2.2.2 Justification of Choice

The Spine-Leaf model was chosen for this project because it reflects the architecture for modern enterprise data centers. It addresses the performance and scalability requirements of a multi-cabinet infrastructure while maintaining the security and zone separation necessary for hosting services across different trust levels, from internal database and application services to externally accessible web services in the DMZ. These choices transform a basic topology into a production-ready, carrier-grade data center blueprint.

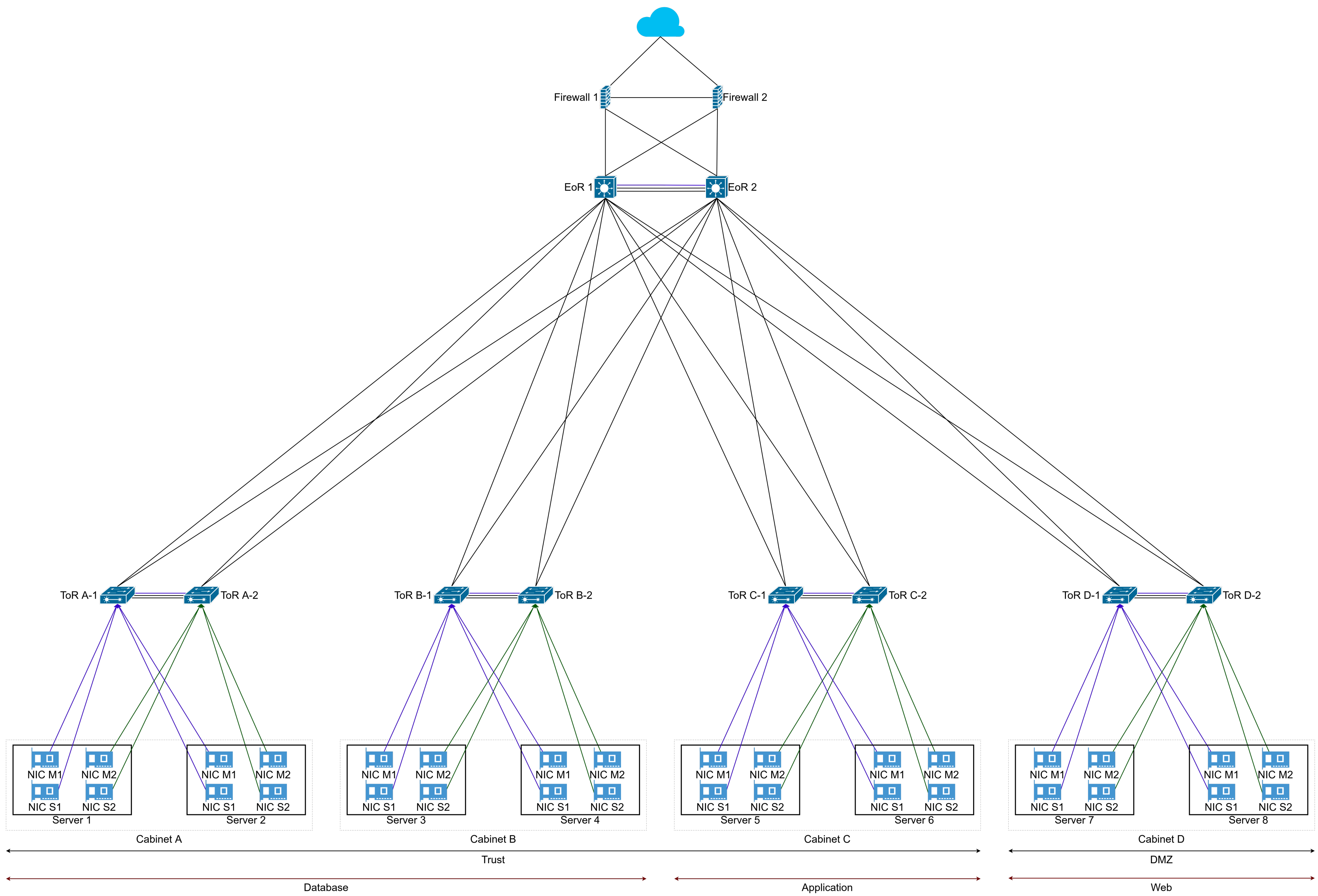


Figure 2.1: *Spine-Leaf topology of the proposed data center.*

2.2.3 Leaf Layer

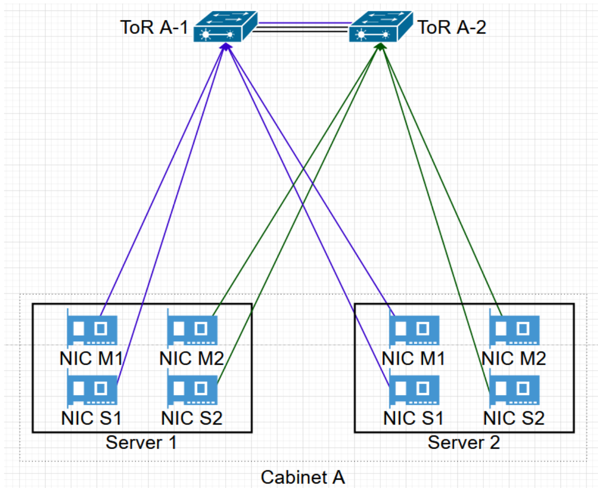


Figure 2.2: *Leaf Layer*

1. Cabinets and Servers: The physical infrastructure consists of four server cabinets (A, B, C, and D). Each cabinet houses two servers, resulting in a total of eight physical servers. Cabinet A contains Servers 1 and 2; Cabinet B contains Servers 3 and 4; Cabinet C contains Servers 5 and 6; Cabinet D contains Servers 7 and 8.
Each server contains two active network cards (M1 and S1) and two backup network cards (M2 and S2):
- M: Management network card, used by the system administrator to access the server for configuration.
- S: Service network card, used for application and production traffic between servers and services.

2. ToR Switches: Each cabinet is served by two ToR switches, one active and one standby, providing local connectivity for the servers within that cabinet. In each cabinet:
- ToR X-1 (active) connects to the two active network cards (M1 and S1) in the two servers in that cabinet.
- ToR X-2 (standby) connects to the two backup network cards (M2 and S2) in the two servers in that cabinet.

ToRs A-Y connect to the two servers in Cabinet A, ToRs B-Y connect to the two servers in Cabinet B, ToRs C-Y connect to the two servers in Cabinet C, ToRs D-Y connect to the two servers in Cabinet D.

2.2.4 Spine Layer

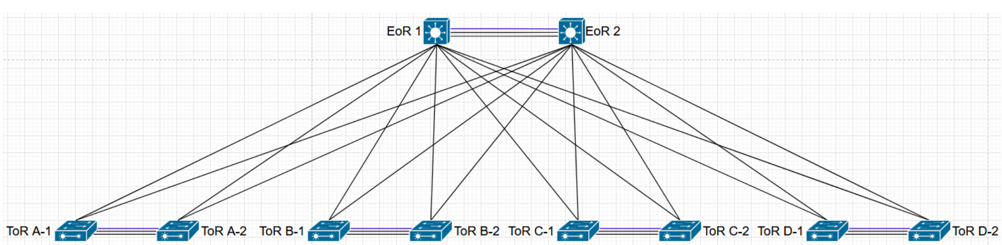


Figure 2.3: *Spine Layer*

EoR Switches: All ToR switches uplink to the core layer, which consists of two End of Row (EoR) switches. These serve as the aggregation point for the entire server farm.
- EoR 1 and EoR 2 are configured in a high-availability (active/standby) pair.
- Each ToR maintains a connection to both EoR 1 and EoR 2. This ensures that if the active path fails, traffic can instantly switch to the standby core switch.
- A direct link exists between EoR 1 and EoR 2 to facilitate synchronization and failover detection (heartbeat).

2.2.5 Network Edge

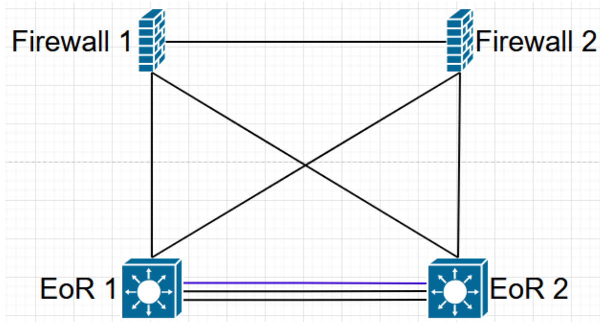


Figure 2.4: *Network Edge*

Firewalls: The core switches connect directly to the network security layer, which is comprised of a pair of firewalls.
- Firewall 1 (Active) and Firewall 2 (Standby) provide north-south security for the server traffic.
- Both EoR 1 and EoR 2 are connected to both Firewall 1 and Firewall 2, eliminating a single point of failure in the uplink path.
- A direct link connects Firewall 1 and Firewall 2 to maintain session state synchronization and monitor health for seamless failover.

2.2.6 Cloud Connection

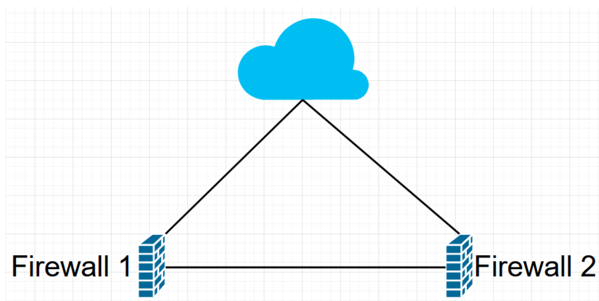


Figure 2.5: *External cloud and network access*

The two perimeter firewalls connect the internal network to the external cloud network. All inbound and outbound traffic between the internal environment and the external network traverses these firewalls, where untrusted traffic is inspected and filtered according to security policies.

2.2.7 Cabinet Structure

The physical architecture is organized into four cabinets, each representing a specific function and security zone:

Table 2.1: *Cabinet organization by zone and function*

Cabinet	Zone	Function	Servers	ToR Pair	VLANs
A	Trust	Database	2 servers	A-1, A-2	110, 120
B	Trust	Database	2 servers	B-1, B-2	110, 120
C	Trust	Application	2 servers	C-1, C-2	110, 130
D	DMZ	Web	2 servers	D-1, D-2	210, 220

Each cabinet contains two servers with four NICs per server and two ToR switches. The ToR switches are configured as an MLAG pair, operating in active-standby mode with one switch active and one standby.

2.3 Requirements Specification

2.3.1 Actors:

The system supports several categories of actors, each with rights and responsibilities adapted to their role. Six primary actors interact with the data center system, as summarized in Table 2.2.

Table 2.2: *Primary actors of the data center system*

Actor	Role and responsibilities
External User	Client accessing web services hosted in the DMZ zone via HTTPS.
IT Administrator	Performs user authentication and monitors cabinet health across all four cabinets.
Network Admin	Configures switches, VLANs, VRRP redundancy, MLAG active-standby pairs, and trunk/access ports.
System Admin	Manages server virtual machines and multi-NIC configuration on Linux servers.
Security Admin	Manages firewall rules and enforces DMZ/Trust zone separation with default-deny policies.
Super Admin	Has full system access and inherits all capabilities from Network, System, Security, and IT Administrators.

2.3.2 Functional Requirements

The data center must fulfill the following functional requirements:
- The system must isolate web servers from application and database servers using firewall-enforced DMZ and Trust zones.
- The system must provide redundant gateway IPs so that server connectivity continues in case of failure.
- The system must support automatic failover between redundant ToRs, EoRs and firewalls with zero packet loss.
- The system must allow IT administrators to remotely configure all network devices via SSH on a dedicated management VLAN.
- The system should be ready for clients to deploy their own applications (web, application, and database) on the infrastructure.

2.3.3 Non-Functional Requirements

Non-functional requirements define the quality expectations and overall constraints of the data center. The main non-functional requirements identified are:
- **Redundancy:** No single point of failure at any layer. Spine, leaf, servers, and firewalls must all have redundancy.
- **Security:** DMZ and Trust zones must be strictly separated by firewall policies with default-deny rules.
- **Scalability:** The architecture must support adding additional cabinets without redesigning the network.
- **Manageability:** All devices must be accessible via SSH on a dedicated management network (VLAN 110 and VLAN 210).

2.3.4 Choice of Modeling Language

We chose UML (Unified Modeling Language) to model our system because of its robustness and wide adoption. This language supports the representation of many kinds of systems, especially IT systems, through its object-oriented approach. UML offers strong expressiveness for capturing requirements from several perspectives; we emphasize the functional view through use case diagrams.

2.3.5 Use Case Description

Figure 2.6 presents the use case diagram with six actors: External User, Network Admin, System Admin, Security Admin, IT Admin, and Super Admin. Use cases cover DMZ access, network configuration (VLAN, VRRP, MLAG), server management, security, and system management. «`include`» arrows require authentication for all configuration actions, while «`extend`» arrows show optional behaviors. Generalization arrows show Super Admin inherits from all other admins.

2.3.6 Use Case Diagram

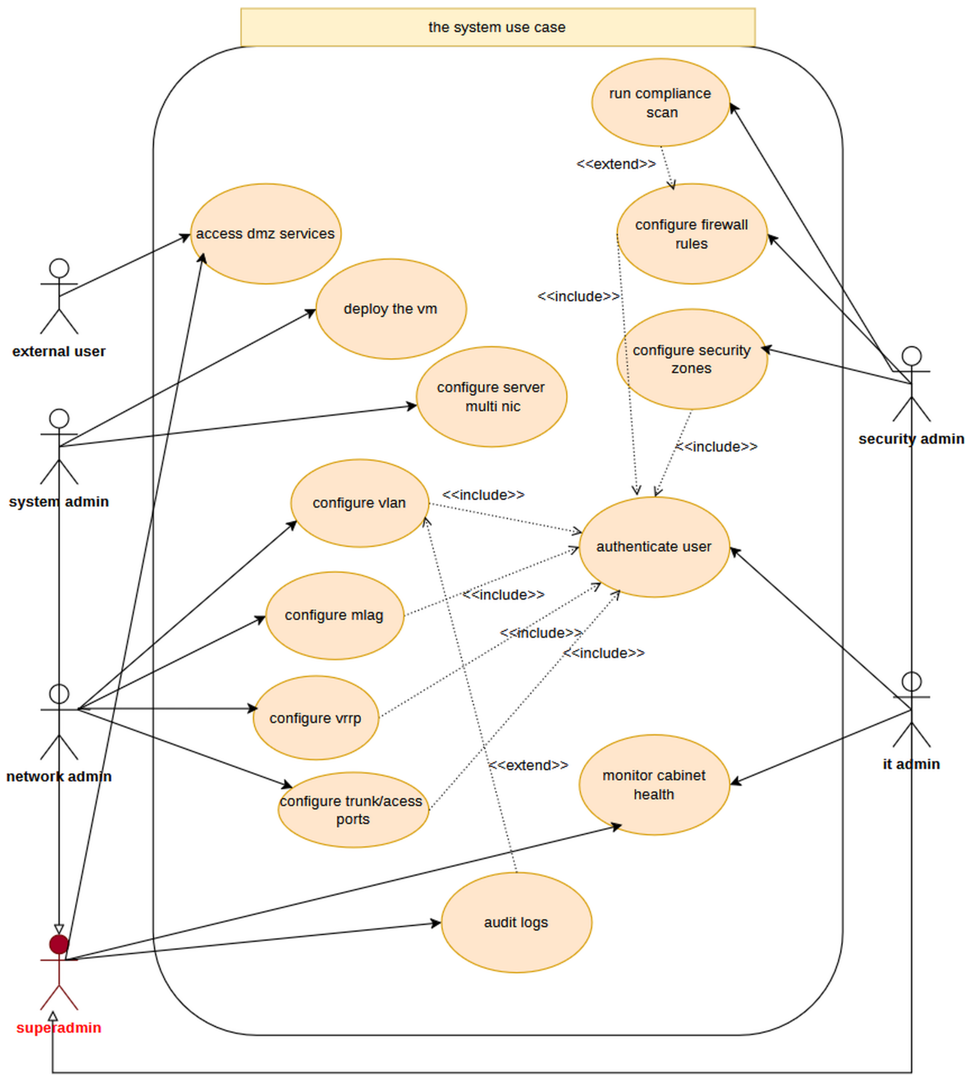


Figure 2.6: *Use case diagram of the data center system.*

The diagram presents all actors interacting with the data center network architecture. Actors are categorized by their access zone and authentication method.

2.4 Sequence Diagrams

2.4.1 Network Administrator Configuring EoR Spine Switches

The sequence diagram in Figure 2.7 illustrates the steps a network administrator follows to configure the EoR spine switches (EoR-1 and EoR-2): initial SSH setup, MLAG peer configuration, and downlink binding to ToR switches per cabinet.

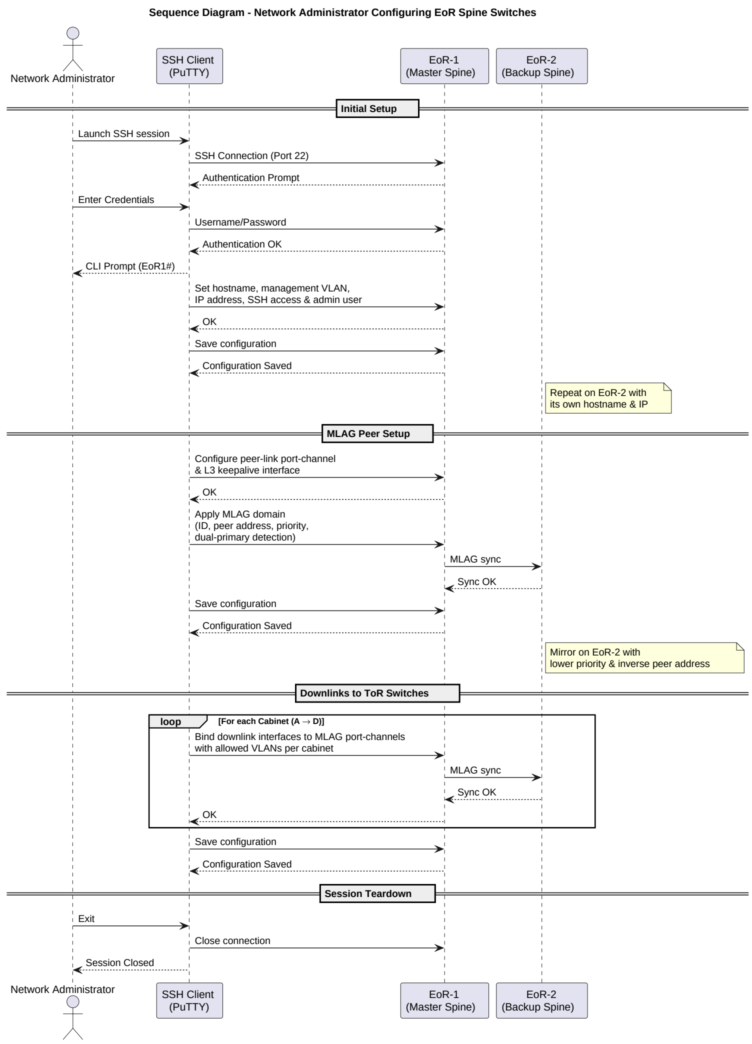


Figure 2.7: *Sequence diagram: Network administrator configuring EoR spine switches.*

2.4.2 System Administrator Configuring a Server

The sequence diagram in Figure 2.8 illustrates the steps a system administrator follows to configure a Linux server: SSH connection, enabling bonding and 802.1Q kernel modules, configuring physical interfaces, creating `bond0` (management, VLAN 110) and `bond1` (service, VLAN 120) with active-backup slaves, and verifying interface reachability before session teardown.

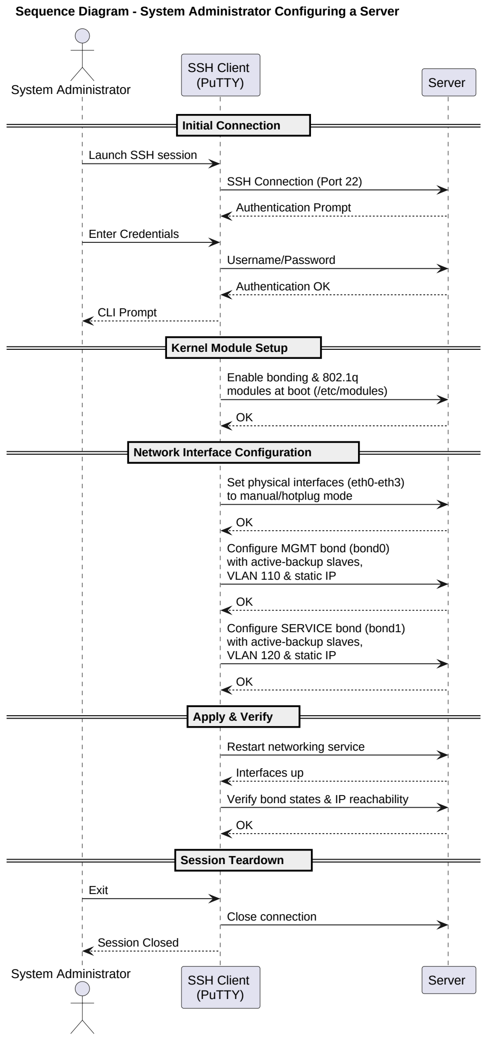


Figure 2.8: *Sequence diagram: System administrator configuring a server.*

2.5 Low-Level Design (LLD)

2.5.1 Device inventory

The data center comprises 2 spine switches, 8 leaf switches, 2 firewalls, and 8 servers. Table 2.3 shows one representative device per type; the full inventory is in Appendix B, Figure B.1.

Table 2.3: *Sample device inventory (one device per type)*

Device	Model/OS	Role	Location
EoR-1	Arista vEOS	Spine switch	Core rack
ToR A-1	Arista vEOS	Leaf switch	Cabinet A
FW-1	pfSense	Firewall	Core rack
Server 1	Linux	Database server	Cabinet A

2.5.2 Port connections

Physical cabling between EoR, ToR, firewall, and server ports is summarized below for one representative path (ToR A-1, EoR-1, FW-1, Server 1). The complete matrices are in Appendix B, Figures B.3–B.5.

Table 2.4: *ToR A-1 → Server 1 (access, copper trunk).*

Source port	Type	Destination	Dest. port	VLAN
Ethernet0/0/1	Access	Server 1	NIC M1	110
Ethernet0/0/2	Access	Server 1	NIC S1	120

Table 2.5: *ToR A-1 → EoR-1 (uplink, fiber trunk).*

Source port	Type	Destination	Dest. port	VLANs
GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/1	110, 120, 130

Table 2.6: *EoR-1 → FW-1 (edge, fiber trunk).*

Source port	Type	Destination	Dest. port	VLANs
GigabitEthernet0/0/24	Trunk	FW-1	GigabitEthernet1/0/0	110, 120, 130, 210, 220

Each ToR uplinks to both EoR switches; both EoRs connect to both firewalls; servers dual-home to their cabinet ToR pair. Trunk ports carry multiple VLANs between switches. Spine-to-leaf, spine-to-spine, and spine-to-firewall trunks carry all five VLANs (110, 120, 130, 210, 220).

2.5.3 VLAN Design

Based on the security zone requirements and traffic separation needs, we defined the following VLANs from the LLD:

	A	B	C	D	E	F	G
1	vlan	Zone	VLAN Name	Subnet	Gateway	Purpose	Devices
2	110	Trust	Management	10.10.10.0/24	10.10.10.254	Device management	All devices
3	120	Trust	Database	10.10.20.0/24	10.10.20.254	Database tier	Servers 1-4
4	130	Trust	App Servers	10.10.30.0/24	10.10.30.254	Application tier	Servers 5, 6
5	210	DMZ	DMZ management	10.20.10.0/24	10.20.10.254	Public web tier	Servers 7, 8
6	220	DMZ	DMZ service	10.20.20.0/24	10.20.20.254	Public web tier	Servers 7, 8

Figure 2.9: *VLAN assignment table.*

All VLANs use /24 subnets (254 usable addresses), providing sufficient capacity for current needs and future expansion.

2.5.4 Server IP Assignments

Each server uses `bond0` (M1+M2) for management traffic and `bond1` (S1+S2) for service traffic. ToR and EoR switches use a management SVI on VLAN 110; firewalls use `lagg0` sub-interfaces per VLAN. Table 2.7 lists one example per device type (IP and interface); the complete plan is in Appendix B, Figure B.2.

Table 2.7: *Sample IP and interface assignments (one device per type).*

Device	V110 (mgmt)		V120 (DB)		V130 (app)		V210 (DMZ mgmt)		V220 (DMZ web)	
	IP	Interface	IP	Interface	IP	Interface	IP	Interface	IP	Interface
Server 1	10.10.10.1	bond0 (M1+M2)	10.10.20.1	bond1 (S1+S2)	n/a	n/a	n/a	n/a	n/a	n/a
ToR A-1	10.10.10.200	SVI Vlan110	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
EoR-1	10.10.10.208	SVI Vlan110	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
FW-1	10.10.10.210	lagg0.110	10.10.20.210	lagg0.120	10.10.30.210	lagg0.130	10.20.10.210	lagg0.210	10.20.20.210	lagg0.220

2.6 Gateway & VRRP Design

Each VLAN uses a Virtual Router Redundancy Protocol (VRRP) virtual IP address as its default gateway. The firewalls participate in VRRP to provide gateway redundancy and seamless failover, eliminating the need for server reconfiguration during a firewall failure or switchover.

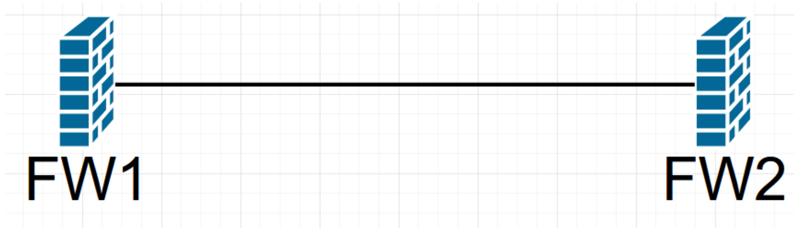


Figure 2.10: *Firewalls VRRP*

2.7 MLAG Design

All switch pairs (ToRs and EoRs) are configured with MLAG setup.
- **Active-standby operation:** One switch forwards traffic while the standby remains ready; on failure, the standby takes over.
- **Sub-second failover:** If the active switch fails, traffic continues through the standby with minimal interruption.

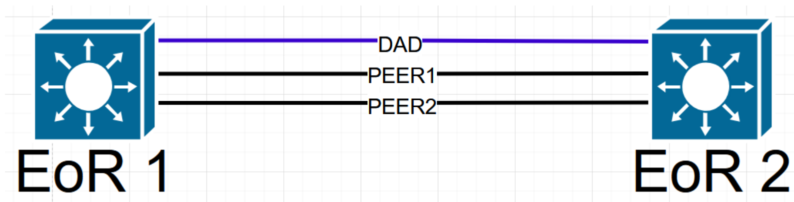


Figure 2.11: *EoRs Pair MLAG Architecture*

2.8 Firewall Rules

Firewall policies enforce strict zone isolation with default-deny rules. Traffic from Untrust to DMZ is permitted only for HTTP and HTTPS. Traffic from DMZ to Trust is permitted only for application-specific ports. All other cross-zone traffic is denied by default. Trust-zone outbound traffic to the external network is permitted via NAT. Management access to all devices is restricted to SSH only.

2.9 Conclusion

This chapter presented a complete spine-leaf data center design with two spines, eight leaves, full-mesh connectivity, and MLAG active-standby pairs at the ToR and EoR layers. Zone-based security separates DMZ, Trust, and Untrust zones via firewall policies, while VRRP supports 99.99% availability targets.

CHAPTER 3

IMPLEMENTATION AND SIMULATION

3.1 Introduction

This chapter implements and validates the data center design from Chapter 2, translating the spine-leaf topology, VLANs, IP addressing, and redundancy into a working simulation. Challenges and their resolutions are documented below.

3.2 Environment

3.2.1 Initial Tool Selection

- Planned stack: Huawei eNSP[12] + VirtualBox[13] for servers.
- Failure: VirtualBox 5.2.44 crashes on Windows 11; eNSP reached end of life in 2022; Hyper-V conflict.

3.2.2 Migration to VMware and PNETLab

- VMware Workstation Pro 17[14] + PNETLab 5.1[15] + PuTTY[16] + Linux Debian servers.
- Resolved compatibility and multi-NIC requirements.

3.2.3 Device Image Issues with Huawei CE12800

- Unstable boot, 4 ports by default, high RAM (8 GB+), no MLAG on available image.

3.2.4 Switch to Arista vEOS

Arista vEOS: reliable start, 24 ports, 2 GB RAM, VLANs, trunks, VRRP, and MLAG. All spine and leaf switches were recreated.

3.3 Device Configurations

3.3.1 PNETLab interface naming (Arista vEOS)

Arista vEOS in PNETLab allows only Ethernet interfaces, while the LLD uses different ports. To overcome this challenge, we used the following port mapping in PNETLab:

Table 3.1: *PNETLab (Arista vEOS) to LLD port mapping.*

Switch	LLD interface(s)	PNETLab interface(s)
ToR	`Ethernet0/0/1–Ethernet0/0/4`	`Eth1–Eth4`
ToR	`GigabitEthernet0/0/1–GigabitEthernet0/0/5`	`Eth5–Eth9`
EoR	`GigabitEthernet0/0/X`	`EthX`

3.3.2 Spine switch (EoR-1)

On EoR-1 we configured: VLANs 110, 120, 130, 210, 220; trunk Port-Channels (Po3–Po10) to all ToR switches; MLAG peer-link Po1 and firewall trunks Po20/Po30; management SVI 10.10.10.208/24 on VLAN 110; MLAG domain `MLAG-EOR` with peer 10.10.10.209. EoR-2 is configured identically to EoR-1; the only difference is MLAG priority, with EoR-2 operating as the secondary peer. Verification screenshots: Appendix E, Figures E.1–E.4.

3.3.3 Leaf switch (ToR A-1)

On ToR A-1 we configured: VLANs 110–220; server Port-Channels Po2–Po5 (mgmt/service bonds); uplink Po6 to both EoRs; MLAG peer-link Po1; management 10.10.10.200/24; MLAG domain `MLAG-TORA` with peer 10.10.10.201. ToR A-2 is configured identically to ToR A-1; the only difference is MLAG priority, with ToR A-2 operating as the secondary peer. Verification: Appendix E, Figures E.5–E.8.

3.3.4 Firewalls (FW-1 and FW-2)

On FW-1/FW-2 we configured: VLAN sub-interfaces on `lagg0` (110–220); per-zone IPs (e.g. 10.10.10.210 on FW-1); CARP virtual gateways (.254) on all VLANs; zone-based rules (DMZ/Trust separation, default deny). FW-1 master, FW-2 backup. Verification: Appendix E, Figures E.9–E.13.

3.3.5 Server (Server 1)

On Server 1 we configured: `bond0` 10.10.10.1/24 (mgmt, eth0+eth2); `bond1` 10.10.20.1/24 (service, eth1+eth3). Servers 2 through 8 are configured identically; only the management and service IP addresses differ according to the LLD. Verification: Appendix E, Figure E.14.

3.4 Connectivity Testing

After all devices were configured, ICMP ping tests were executed between Linux servers to validate end-to-end connectivity. Table 3.2 lists every test with its scope (intra/inter cabinet, VLAN, management or service plane, and security zone) and the observed result. Terminal screenshots are provided in Appendix D.

Table 3.2: *Server-to-server ping test results*

Source	Destination	Cabinet	VLAN	Plane	Zone	Result
Server 1	Server 2 (10.10.10.2)	A	110	Mgmt	Trust	ALLOWED
Server 1	Server 2 (10.10.20.2)	A	120	Service	Trust	ALLOWED
Server 1	Server 3 (10.10.20.3)	A–B	120	Service	Trust	ALLOWED
Server 1	Server 5 (10.10.30.1)	A–C	120–130	Service	Trust	ALLOWED
Server 7	Server 8 (10.20.20.2)	D	220	Service	DMZ	ALLOWED
Server 1	Server 7 (10.20.10.1)	A–D	110–210	Mgmt	Trust–DMZ	ALLOWED
Server 7	Server 1 (10.10.10.1)	D–A	210–110	Mgmt	DMZ–Trust	BLOCKED

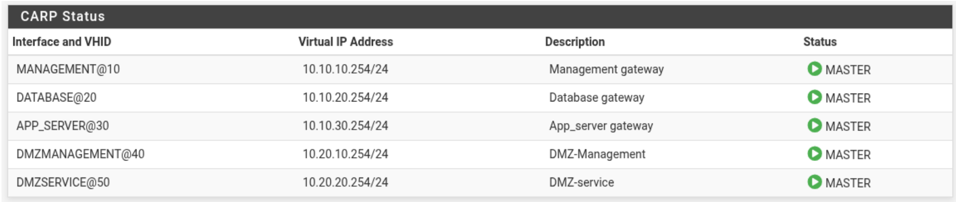
All `ALLOWED` flows completed successfully, confirming correct VLAN assignment, inter-VLAN routing on the EoR switches, and server bonding. The single `BLOCKED` test (DMZ–Trust management) validates zone-based firewall policy. Together, the tests cover intra- and inter-cabinet paths, management and service planes, and both intra-zone and inter-zone traffic.

3.5 Redundancy & Failure Testing

Redundancy was validated by shutting down the active node in each redundant pair and verifying that the standby assumed control. Firewall and EoR pairs were each tested once. At the leaf layer, all four ToR pairs (Cabinets A–D) use the same MLAG active-standby design; the ToR B test below is documented as representative, because the failover procedure and expected outcome are identical for every cabinet pair.

3.5.1 VRRP (CARP) firewall failover test

- Stop FW-1; keep pings to CARP gateway addresses.
- Observe FW-2 `MASTER` on all VLANs (Figure 3.1).
- Result: active-standby firewall failover confirmed.



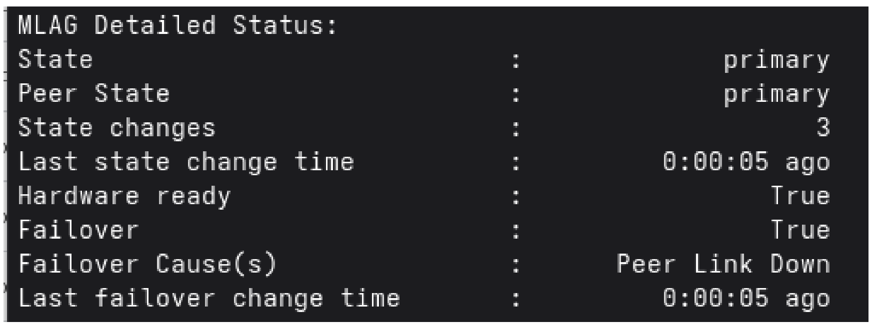
CARP Status

Interface and VHID	Virtual IP Address	Description	Status
MANAGEMENT@10	10.10.10.254/24	Management gateway	MASTER
DATABASE@20	10.10.20.254/24	Database gateway	MASTER
APP_SERVER@30	10.10.30.254/24	App_server gateway	MASTER
DMZMANAGEMENT@40	10.20.10.254/24	DMZ-Management	MASTER
DMZSERVICE@50	10.20.20.254/24	DMZ-service	MASTER

Figure 3.1: *VRRP (CARP) on FW-2 after FW-1 failure.*

3.5.2 EoR MLAG failover test

- Shut down EoR-1; run `show mlag detail` on EoR-2.
- Observe `Failover:  True`, cause `Peer Link Down` (Figure 3.2).
- Result: backup spine continues forwarding.

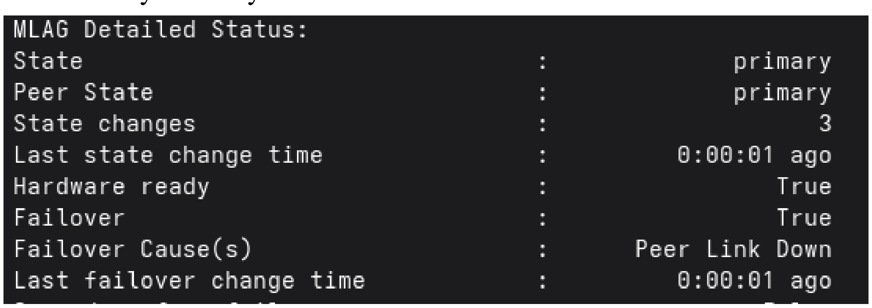


```
MLAG Detailed Status:
State                            :          primary
Peer State                       :          primary
State changes                    :                3
Last state change time           :     0:00:05 ago
Hardware ready                   :             True
Failover                         :             True
Failover Cause(s)                :   Peer Link Down
Last failover change time        :     0:00:05 ago
```

Figure 3.2: *MLAG status on EoR-2 after EoR-1 failure.*

3.5.3 ToR B MLAG failover test

- Shut down ToR B-1 with traffic on Cabinet B servers.
- Observe ToR B-2 MLAG primary after peer-link loss (Figure 3.3).
- Result: leaf active-standby recovery confirmed.



```
MLAG Detailed Status:
State                                 :          primary
Peer State                            :          primary
State changes                         :                3
Last state change time                :     0:00:01 ago
Hardware ready                        :             True
Failover                              :             True
Failover Cause(s)                     :   Peer Link Down
Last failover change time             :     0:00:01 ago
```

Figure 3.3: *MLAG status on ToR B-2 after ToR B-1 failure.*

3.6 Simulation Environment Status

The final simulation environment runs on:

- **Host:** Windows 11 Pro with VMware Workstation Pro 17
- **PNETLab VM:** 10 vCPUs, 24GB RAM, 100GB storage
- **Total device count:**

  - 2 spine switches (Arista vEOS)
  - 8 leaf switches (Arista vEOS)
  - 2 firewalls (pfSense)
  - 8 servers (Linux Debian)
- **Resource utilization:** 30% CPU, 70% RAM

All devices start reliably and maintain stable connections.

3.7 Conclusion

This chapter has presented the complete implementation and validation of the data center design. The implemented data center provides a solid foundation for client application deployment, with configuration verification in Appendix E, connectivity tests in Appendix D, and LLD tables in Appendix B.

GENERAL CONCLUSION

This dissertation addressed the design and simulation of a modern data center network architecture using spine-leaf topology with full mesh redundancy, active-standby MLAG operation, and zone-based security. The work was structured into three complementary chapters. Chapter 1 established the theoretical foundations, comparing traditional three-tier architecture with modern spine-leaf. Chapter 2 translated these concepts into a concrete design comprising four cabinets, eight ToR leaf switches, two EoR spine switches, and two firewalls in full mesh topology, with five VLANs across Trust and DMZ zones. Chapter 3 implemented the complete design in simulation, migrating from eNSP to PNETLab, from Huawei CE12800 to Arista vEOS, and from VPCS to Linux servers.

The principal achievements of this project are: a complete spine-leaf implementation with 2 spines, 8 leaves, 16 paths, and full mesh connectivity; active-standby redundancy at the ToR, EoR, and firewall layers (active-standby with VRRP/MLAG) providing sub-second failover; zone-based security ensuring that web layer compromise cannot propagate to database servers; comprehensive validation through five failure scenarios (spine failure, leaf failure, link failure, NIC failure, and cabinet failure); professional documentation including HLD, LLD, UML diagrams, configuration templates, and test results; and thorough problem-solving documentation of the migration from incompatible tools.

Despite these achievements, several limitations remain. Complete cabinet failure isolates all servers in that cabinet; true high availability would require cross-cabinet replication. Storage networking (SAN/NAS) was not implemented, and the design remains simulated rather than deployed on physical hardware. These limitations inform future work directions.

In conclusion, this project successfully demonstrates a production-ready, carrier-grade data center blueprint that is validated, documented, and ready for enterprise deployment.

APPENDIX A

DEFINITIONS

A.1 Network Architecture Definitions

Spine Switch

A **spine switch** forms the core forwarding layer in spine-leaf architecture. It connects only to leaf switches (no direct server connections) and operates at Layer 3 for routing between leaf switches. In our design, EoR-1 and EoR-2 serve as spine switches.

Leaf Switch (ToR)

A **leaf switch** (Top of Rack) provides server connectivity in the access layer. It connects to servers via access ports and uplinks to all spine switches via trunk ports. In our design, we have eight leaf switches (ToR A-1 through ToR D-2).

Spine-Leaf Architecture

Spine-Leaf architecture is a two-tier network topology where every leaf switch connects to every spine switch in a full mesh. Any server reaches any other server in exactly two hops, providing consistent latency and ECMP load balancing.

Three-Tier Architecture

Three-Tier architecture is a traditional data center architecture consisting of core layer (backbone), aggregation layer (policy enforcement), and access layer (server connectivity). Designed for north-south traffic.

EoR (End of Row)

An **End-of-Row switch** is placed at the end of a row of server racks, connecting all servers in that row. In our design, EoR-1 and EoR-2 function as spine switches.

ToR (Top of Rack)

A **Top-of-Rack switch** is located at the top of a server cabinet, connecting all servers within that cabinet. In our design, we have eight ToR switches (two per cabinet).

Full Mesh

Full mesh is a topology where every device connects directly to every other device. In spine-leaf architecture, every leaf switch connects to every spine switch.

Cluster

A **cluster** is a group of physical servers that act as a single logical unit to provide high availability and load balancing. In our project, one Linux instance represents a cluster of four physical servers.

A.2 VLAN and Switching Definitions

VLAN (Virtual Local Area Network)

A **VLAN** is a logical subdivision of a physical network that groups devices based on function or security requirements rather than physical location. VLANs isolate traffic, reduce broadcast domains, and improve security.

Trunk Port

A **trunk port** is a switch port that carries traffic for multiple VLANs simultaneously using 802.1Q tagging. Used for inter-switch connections.

Access Port

An **access port** is a switch port that belongs to a single VLAN and carries untagged traffic. Used for connecting end devices like servers.

A.3 Redundancy and High Availability Definitions

VRRP (Virtual Router Redundancy Protocol)

VRRP is a protocol that allows multiple routers to share a single virtual IP address. If the master router fails, a backup takes over automatically within seconds, providing gateway redundancy.

MLAG (Multi-Chassis Link Aggregation)

MLAG is a technology that allows two physical switches to appear as a single logical switch to connected devices. In our design, MLAG pairs operate in active-standby mode with master/backup switches.

Active-Active

Active-active is an alternative redundancy mode where both devices forward traffic simultaneously. This mode is not used in our design.

Active-Standby

Active-standby is a redundancy mode where one device (master) handles all traffic while the other (standby) remains idle until the master fails. VRRP uses this mode for gateways and firewalls. MLAG pairs at the ToR and EoR layers also operate in active-standby mode in our design.

Failover

Failover is the automatic switching from a failed primary component to a redundant backup component.

A.4 Security Definitions

DMZ (Demilitarized Zone)

DMZ is a semi-trusted zone for public-facing services such as web servers. It sits between the internet (untrust) and the internal network (trust). Accessible from internet on specific ports (HTTP/HTTPS).

Trust Zone

The **Trust zone** is the most secure zone for internal services including application servers, database servers, and management networks. No direct internet access.

Untrust Zone

The **Untrust zone** represents all external networks (internet) and is considered completely untrusted. All traffic must pass through firewall inspection.

Firewall

A **firewall** is a security device that inspects and filters network traffic based on configured rules. Enforces zone-based security policies and performs stateful packet inspection.

NAT (Network Address Translation)

NAT is a method that maps private IP addresses to public IP addresses, allowing internal devices to communicate with the internet while hiding internal network structure.

A.5 Virtualization Definitions

Hypervisor

A **hypervisor** is software that creates and runs virtual machines by abstracting the underlying physical hardware (e.g., VMware ESXi).

Virtual Machine (VM)

A **virtual machine** is an emulation of a computer system that runs its own operating system on top of a hypervisor.

Server Cluster

A **server cluster** is a group of physical servers that act as a single logical unit for high availability and load balancing. In our project, one Linux instance represents a cluster.

VMware

VMware is a virtualization platform that allows running multiple virtual machines on a single physical server.

A.6 Simulation and Tools Definitions

eNSP (Enterprise Network Simulation Platform)

eNSP is Huawei's official network simulation platform for designing and testing networks without physical hardware. Requires VirtualBox 5.2.44.

PNETLab (Practical Networking Lab)

PNETLab is a community-maintained network simulation platform (EVE-NG fork) that runs on VMware and supports multiple device images (Arista vEOS, Debian Linux, Huawei CE12800).

Arista vEOS

Arista vEOS (Virtual Extensible Operating System) is Arista's virtual network device emulator used in our project for spine and leaf switches. Supports VLANs, trunk ports, VRRP, OSPF, and MLAG with 2GB RAM per instance.

VMware Workstation Pro

VMware Workstation Pro is a Type-2 hypervisor used to host the PNETLab virtual machine, providing hardware virtualization support.

PuTTY

PuTTY is a free SSH and Telnet client used in our project to access network devices and servers for configuration and testing.

VPCS (Virtual PC Simulator)

VPCS is a lightweight tool included with PNETLab that simulates a simple PC. Limited to one IP address per device, replaced by Linux Tiny Core.

A.7 Server and Network Interface Definitions

NIC (Network Interface Card)

A **NIC** (Network Interface Card) is a hardware component that connects a server to a network. In our design, each server has four NICs (M1, M2, S1, S2).

Management Network (M1/M2)

The **management network** is the network interface used by administrators to access servers for configuration, monitoring, and maintenance. Isolated from service traffic.

Service Network (S1/S2)

The **service network** is the network interface used for application traffic between servers (web to app, app to database). Separate from management network for security.

A.8 Documentation Definitions

HLD (High-Level Design)

HLD (High-Level Design) is a top-level design document showing the overall architecture, major components, and their relationships without implementation details.

LLD (Low-Level Design)

LLD (Low-Level Design) is a detailed design document containing specific IP addresses, VLAN assignments, port connections, configuration commands, and implementation procedures.

UML (Unified Modeling Language)

UML (Unified Modeling Language) is a standardized modeling language used to visualize system design. In our project, we used use case diagrams and sequence diagrams.

Use Case Diagram

A **use case diagram** is a UML diagram showing actors (users or systems) and their interactions with the system. Used to identify functional requirements.

Sequence Diagram

A **sequence diagram** is a UML diagram showing the temporal order of message exchanges between objects. Used to illustrate operational flows and failure scenarios.

APPENDIX B

APPENDIX B: LOW-LEVEL DESIGN (LLD)

B.1 Device inventory

	A	B	C	D
1	Device	Model/OS	Role	Location
2	EoR-1	Arista vEOS	Spine Switch	Core Rack
3	EoR-2	Arista vEOS	Spine Switch	Core Rack
4	ToR-A-1	Arista vEOS	Leaf Switch	Cabinet A
5	ToR-A-2	Arista vEOS	Leaf Switch	Cabinet A
6	ToR-B-1	Arista vEOS	Leaf Switch	Cabinet B
7	ToR-B-2	Arista vEOS	Leaf Switch	Cabinet B
8	ToR-C-1	Arista vEOS	Leaf Switch	Cabinet C
9	ToR-C-2	Arista vEOS	Leaf Switch	Cabinet C
10	ToR-D-1	Arista vEOS	Leaf Switch	Cabinet D
11	ToR-D-2	Arista vEOS	Leaf Switch	Cabinet D
12	FW-1	pfSense	Firewall	Core Rack
13	FW-2	pfSense	Firewall	Core Rack
14	Server 1	Linux Debian	Database Server	Cabinet A
15	Server 2	Linux Debian	Database Server	Cabinet A
16	Server 3	Linux Debian	Database Server	Cabinet B
17	Server 4	Linux Debian	Database Server	Cabinet B
18	Server 5	Linux Debian	App Server	Cabinet C
19	Server 6	Linux Debian	App Server	Cabinet C
20	Server 7	Linux Debian	Web Server	Cabinet D
21	Server 8	Linux Debian	Web Server	Cabinet D

Figure B.1: *Complete device inventory (LLD).*

B.2 IP addressing plan

Device	Mangement (VLAN 110)		DB service (VLAN 120) IP address Interface		APP service (VLAN 130)		DMZ managment VLAN 210		DMZ service (VLAN 220)	
	IP address	Interface	IP address	Interface	IP address	Interface	IP address	Interface	IP address	Interface
DB Server 1 (Server 1)	10.10.10.1	bond0 (NIC M1 + M2)	10.10.20.1	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A	N/A	N/A
DB Server 2 (Server 2)	10.10.10.2	bond0 (NIC M1 + M2)	10.10.20.2	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A	N/A	N/A
DB Server 3 (Server 3)	10.10.10.3	bond0 (NIC M1 + M2)	10.10.20.3	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A	N/A	N/A
DB Server 4 (Server 4)	10.10.10.4	bond0 (NIC M1 + M2)	10.10.20.4	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A	N/A	N/A
App Server 1 (Server 5)	10.10.10.5	bond0 (NIC M1 + M2)	N/A	N/A	10.10.30.1	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A
App Server 2 (Server 6)	10.10.10.6	bond0 (NIC M1 + M2)	N/A	N/A	10.10.30.2	bond1 (NIC S1 + S2)	N/A	N/A	N/A	N/A
Web Server 1 (Server 7)	N/A	N/A	N/A	N/A	N/A	N/A	10.20.10.1	bond0 (NIC M1 + M2)	10.20.20.1	bond1 (NIC S1 + S2)
Web Server 2 (Server 8)	N/A	N/A	N/A	N/A	N/A	N/A	10.20.10.2	bond0 (NIC M1 + M2)	10.20.20.2	bond1 (NIC S1 + S2)
ToR A-1	10.10.10.200	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR A-2	10.10.10.201	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR B-1	10.10.10.202	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR B-2	10.10.10.203	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR C-1	10.10.10.204	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR C-2	10.10.10.205	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR D-1	10.10.10.206	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ToR D-2	10.10.10.207	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
EoR 1	10.10.10.208	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
EoR 2	10.10.10.209	SVI Vlan110	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Firewall 1	10.10.10.210	lagg0.110	10.10.20.210	lagg0.120	10.10.30.210	lagg0.130	10.20.10.210	lagg0.210	10.20.20.210	lagg0.220
Firewall 2	10.10.10.211	lagg0.110	10.10.20.211	lagg0.120	10.10.30.211	lagg0.130	10.20.10.211	lagg0.210	10.20.20.211	lagg0.220

Figure B.2: *Complete IP addressing plan (LLD).*

B.3 Physical port connections

Source Device	Source Port	Port Type	Destination Device	Destination Port	VLAN Mode / Subnet	Cable Type	
ToR-A-1	Ethernet0/0/1	Access	Server 1	Ethernet NIC M1	VLAN 110	Copper	
ToR-A-1	Ethernet0/0/2	Access	Server 1	Ethernet NIC S1	VLAN 120	Copper	
ToR-A-1	Ethernet0/0/3	Access	Server 2	Ethernet NIC M1	VLAN 110	Copper	
ToR-A-1	Ethernet0/0/4	Access	Server 2	Ethernet NIC S1	VLAN 120	Copper	
ToR-A-1	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
ToR-A-1	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
ToR-A-1	GigabitEthernet0/0/3	Trunk	ToR-A-2	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-A-1	GigabitEthernet0/0/4	Trunk	ToR-A-2	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-A-1	GigabitEthernet0/0/5	Trunk	ToR-A-2	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-A-2	Ethernet0/0/1	Access	Server 1	Ethernet NIC M2	VLAN 110	Copper	
ToR-A-2	Ethernet0/0/2	Access	Server 1	Ethernet NIC S2	VLAN 120	Copper	
ToR-A-2	Ethernet0/0/3	Access	Server 2	Ethernet NIC M2	VLAN 110	Copper	
ToR-A-2	Ethernet0/0/4	Access	Server 2	Ethernet NIC S2	VLAN 120	Copper	
ToR-A-2	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
ToR-A-2	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
ToR-A-2	GigabitEthernet0/0/3	Trunk	ToR-A-1	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-A-2	GigabitEthernet0/0/4	Trunk	ToR-A-1	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-A-2	GigabitEthernet0/0/5	Trunk	ToR-A-1	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-1	Ethernet0/0/1	Access	Server 3	Ethernet NIC M1	VLAN 110	Copper	
ToR-B-1	Ethernet0/0/2	Access	Server 3	Ethernet NIC S1	VLAN 120	Copper	
ToR-B-1	Ethernet0/0/3	Access	Server 4	Ethernet NIC M1	VLAN 110	Copper	
ToR-B-1	Ethernet0/0/4	Access	Server 4	Ethernet NIC S1	VLAN 120	Copper	
ToR-B-1	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	
ToR-B-1	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	
ToR-B-1	GigabitEthernet0/0/3	Trunk	ToR-B-2	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-1	GigabitEthernet0/0/4	Trunk	ToR-B-2	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-1	GigabitEthernet0/0/5	Trunk	ToR-B-2	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-2	Ethernet0/0/1	Access	Server 3	Ethernet NIC M2	VLAN 110	Copper	
ToR-B-2	Ethernet0/0/2	Access	Server 3	Ethernet NIC S2	VLAN 120	Copper	
ToR-B-2	Ethernet0/0/3	Access	Server 4	Ethernet NIC M2	VLAN 110	Copper	
ToR-B-2	Ethernet0/0/4	Access	Server 4	Ethernet NIC S2	VLAN 120	Copper	
ToR-B-2	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	
ToR-B-2	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	
ToR-B-2	GigabitEthernet0/0/3	Trunk	ToR-B-1	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-2	GigabitEthernet0/0/4	Trunk	ToR-B-1	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
ToR-B-2	GigabitEthernet0/0/5	Trunk	ToR-B-1	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag

Figure B.3: *Port connections matrix (part 1).*

42	ToR-C-1	Ethernet0/0/1	Access	Server 5	Ethernet NIC M1	VLAN 110	Copper	
43	ToR-C-1	Ethernet0/0/2	Access	Server 5	Ethernet NIC S1	VLAN 130	Copper	
44	ToR-C-1	Ethernet0/0/3	Access	Server 6	Ethernet NIC M1	VLAN 110	Copper	
45	ToR-C-1	Ethernet0/0/4	Access	Server 6	Ethernet NIC S1	VLAN 130	Copper	
46	ToR-C-1	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	
47	ToR-C-1	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	
48	ToR-C-1	GigabitEthernet0/0/3	Trunk	ToR-C-2	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
49	ToR-C-1	GigabitEthernet0/0/4	Trunk	ToR-C-2	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
50	ToR-C-1	GigabitEthernet0/0/5	Trunk	ToR-C-2	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
51								
52	ToR-C-2	Ethernet0/0/1	Access	Server 5	Ethernet NIC M2	VLAN 110	Copper	
53	ToR-C-2	Ethernet0/0/2	Access	Server 5	Ethernet NIC S2	VLAN 130	Copper	
54	ToR-C-2	Ethernet0/0/3	Access	Server 6	Ethernet NIC M2	VLAN 110	Copper	
55	ToR-C-2	Ethernet0/0/4	Access	Server 6	Ethernet NIC S2	VLAN 130	Copper	
56	ToR-C-2	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/6	Allow : VLANs 110 120 130	Fiber	
57	ToR-C-2	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/6	Allow : VLANs 110 120 130	Fiber	
58	ToR-C-2	GigabitEthernet0/0/3	Trunk	ToR-C-1	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
59	ToR-C-2	GigabitEthernet0/0/4	Trunk	ToR-C-1	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
60	ToR-C-2	GigabitEthernet0/0/5	Trunk	ToR-C-1	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
61								
62	ToR-D-1	Ethernet0/0/1	Access	Server 7	Ethernet NIC M1	VLAN 210	Copper	
63	ToR-D-1	Ethernet0/0/2	Access	Server 7	Ethernet NIC S1	VLAN 220	Copper	
64	ToR-D-1	Ethernet0/0/3	Access	Server 8	Ethernet NIC M1	VLAN 210	Copper	
65	ToR-D-1	Ethernet0/0/4	Access	Server 8	Ethernet NIC S1	VLAN 220	Copper	
66	ToR-D-1	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/7	Allow : VLANs 210 220	Fiber	
67	ToR-D-1	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/7	Allow : VLANs 210 220	Fiber	
68	ToR-D-1	GigabitEthernet0/0/3	Trunk	ToR-D-2	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
69	ToR-D-1	GigabitEthernet0/0/4	Trunk	ToR-D-2	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
70	ToR-D-1	GigabitEthernet0/0/5	Trunk	ToR-D-2	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag
71								
72	ToR-D-2	Ethernet0/0/1	Access	Server 7	Ethernet NIC M2	VLAN 210	Copper	
73	ToR-D-2	Ethernet0/0/2	Access	Server 7	Ethernet NIC S2	VLAN 220	Copper	
74	ToR-D-2	Ethernet0/0/3	Access	Server 8	Ethernet NIC M2	VLAN 210	Copper	
75	ToR-D-2	Ethernet0/0/4	Access	Server 8	Ethernet NIC S2	VLAN 220	Copper	
76	ToR-D-2	GigabitEthernet0/0/1	Trunk	EoR-1	GigabitEthernet0/0/8	Allow : VLANs 210 220	Fiber	
77	ToR-D-2	GigabitEthernet0/0/2	Trunk	EoR-2	GigabitEthernet0/0/8	Allow : VLANs 210 220	Fiber	
78	ToR-D-2	GigabitEthernet0/0/3	Trunk	ToR-D-1	GigabitEthernet0/0/3	Allow : VLANs 110 120 130	Fiber	M-Lag
79	ToR-D-2	GigabitEthernet0/0/4	Trunk	ToR-D-1	GigabitEthernet0/0/4	Allow : VLANs 110 120 130	Fiber	M-Lag
80	ToR-D-2	GigabitEthernet0/0/5	Trunk	ToR-D-1	GigabitEthernet0/0/5	Allow : VLANs 110 120 130	Fiber	M-Lag

Figure B.4: *Port connections matrix (part 2).*

82	EoR-1	GigabitEthernet0/0/1	Trunk	ToR-A-1	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
83	EoR-1	GigabitEthernet0/0/2	Trunk	ToR-A-2	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
84	EoR-1	GigabitEthernet0/0/3	Trunk	ToR-B-1	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
85	EoR-1	GigabitEthernet0/0/4	Trunk	ToR-B-2	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
86	EoR-1	GigabitEthernet0/0/5	Trunk	ToR-C-1	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
87	EoR-1	GigabitEthernet0/0/6	Trunk	ToR-C-2	GigabitEthernet0/0/1	Allow : VLANs 110 120 130	Fiber	
88	EoR-1	GigabitEthernet0/0/7	Trunk	ToR-D-1	GigabitEthernet0/0/1	Allow : VLANs 210 220	Fiber	
89	EoR-1	GigabitEthernet0/0/8	Trunk	ToR-D-2	GigabitEthernet0/0/1	Allow : VLANs 210 220	Fiber	
90	EoR-1	GigabitEthernet0/0/21	Trunk	EoR-2	GigabitEthernet0/0/21		Fiber	M-Lag
91	EoR-1	GigabitEthernet0/0/22	Trunk	EoR-2	GigabitEthernet0/0/22		Fiber	M-Lag
92	EoR-1	GigabitEthernet0/0/23	Trunk	EoR-2	GigabitEthernet0/0/23		Fiber	M-Lag
93	EoR-1	GigabitEthernet0/0/24	Trunk	FW-1	GigabitEthernet1/0/0		Fiber	
94	EoR-1	GigabitEthernet0/0/25	Trunk	FW-2	GigabitEthernet1/0/0		Fiber	
95								
96	EoR-2	GigabitEthernet0/0/1	Trunk	ToR-A-1	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
97	EoR-2	GigabitEthernet0/0/2	Trunk	ToR-A-2	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
98	EoR-2	GigabitEthernet0/0/3	Trunk	ToR-B-1	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
99	EoR-2	GigabitEthernet0/0/4	Trunk	ToR-B-2	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
100	EoR-2	GigabitEthernet0/0/5	Trunk	ToR-C-1	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
101	EoR-2	GigabitEthernet0/0/6	Trunk	ToR-C-2	GigabitEthernet0/0/2	Allow : VLANs 110 120 130	Fiber	
102	EoR-2	GigabitEthernet0/0/7	Trunk	ToR-D-1	GigabitEthernet0/0/2	Allow : VLANs 210 220	Fiber	
103	EoR-2	GigabitEthernet0/0/8	Trunk	ToR-D-2	GigabitEthernet0/0/2	Allow : VLANs 210 220	Fiber	
104	EoR-2	GigabitEthernet0/0/21	Trunk	EoR-1	GigabitEthernet0/0/21		Fiber	M-Lag
105	EoR-2	GigabitEthernet0/0/22	Trunk	EoR-1	GigabitEthernet0/0/22		Fiber	M-Lag
106	EoR-2	GigabitEthernet0/0/23	Trunk	EoR-1	GigabitEthernet0/0/23		Fiber	M-Lag
107	EoR-2	GigabitEthernet0/0/24	Trunk	FW-1	GigabitEthernet1/0/1		Fiber	
108	EoR-2	GigabitEthernet0/0/25	Trunk	FW-2	GigabitEthernet1/0/1		Fiber	
109								
110	FW-1	GigabitEthernet1/0/2	Trunk	FW-2	GigabitEthernet1/0/2		Fiber	VRRP

Figure B.5: *Port connections matrix (part 3).*

APPENDIX C

APPENDIX C: DEVICE CONFIGURATIONS AND SCRIPTS

C.1 Server network configurations

The following listings show the `/etc/network/interfaces` configuration applied on each Linux server (Servers 1–8), in cabinet order.

C.1.1 Server 1

Listing C.1: *Server 1: network interfaces (Cabinet A).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.1
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
```

```
iface bond1 inet static
    address 10.10.20.1
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.2 Server 2

Listing C.2: *Server 2: network interfaces (Cabinet A).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.2
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.10.20.2
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.3 Server 3

Listing C.3: *Server 3: network interfaces (Cabinet B).*

```
source /etc/network/interfaces.d/*
```

```
auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.3
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.10.20.3
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.4 Server 4

Listing C.4: *Server 4: network interfaces (Cabinet B).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual
```

```
allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.4
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.10.20.4
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.5 Server 5

Listing C.5: *Server 5: network interfaces (Cabinet C).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.5
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

```
auto bond1
iface bond1 inet static
    address 10.10.30.1
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.6 Server 6

Listing C.6: *Server 6: network interfaces (Cabinet C).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.10.10.6
    netmask 255.255.255.0
    gateway 10.10.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.10.30.2
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.7 Server 7

Listing C.7: *Server 7: network interfaces (Cabinet D, DMZ).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.20.10.1
    netmask 255.255.255.0
    gateway 10.20.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.20.20.1
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.1.8 Server 8

Listing C.8: *Server 8: network interfaces (Cabinet D, DMZ).*

```
source /etc/network/interfaces.d/*

auto lo
iface lo inet loopback

allow-hotplug eth0
iface eth0 inet manual

allow-hotplug eth1
iface eth1 inet manual

allow-hotplug eth2
```

```
iface eth2 inet manual

allow-hotplug eth3
iface eth3 inet manual

auto bond0
iface bond0 inet static
    address 10.20.10.2
    netmask 255.255.255.0
    gateway 10.20.10.254
    bond-slaves eth0 eth2
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200

auto bond1
iface bond1 inet static
    address 10.20.20.2
    netmask 255.255.255.0
    bond-slaves eth1 eth3
    bond-mode active-backup
    bond-miimon 100
    bond-updelay 200
    bond-downdelay 200
```

C.2 Firewall configuration (FW-1 and FW-2)

Both firewalls run pfSense and are configured identically except for their individual interface IPs. Configuration is performed through the pfSense web GUI and follows seven sequential phases.

C.2.1 LAGG interface creation

Navigate to `Interfaces` → `LAGGs` → `Add` and configure the link aggregation group:

- Parent interfaces: `vtnet0` and `vtnet1`
- LAGG protocol: LACP
- LACP timeout: Slow
- Hash algorithm: Layer 2/3/4
- Description: `Uplink-to-EoR`

C.2.2 VLAN creation

Navigate to `Interfaces` → `VLANs` → `Add` and create five VLAN sub-interfaces on `lagg0`:

Table C.1: *VLAN sub-interfaces on* `lagg0`*.*

Parent	VLAN tag	Description
lagg0	110	Management
lagg0	120	Database
lagg0	130	App
lagg0	210	DMZ-Management
lagg0	220	DMZ-Service

C.2.3 Interface assignment

Navigate to `Interfaces` → `Interface Assignments` and add each sub-interface in order:

Table C.2: *Firewall interface assignments.*

Sub-interface	Assigned name
lagg0.110	OPT1
lagg0.120	OPT2
lagg0.130	OPT3
lagg0.210	OPT4
lagg0.220	OPT5
vtnet2	OPT6

C.2.4 Interface IP configuration

Navigate to `Interfaces` → `OPTx` for each interface and apply the following static IPv4 addresses. All interfaces are enabled with IPv4 configuration type set to Static.

Table C.3: *Firewall interface IP addresses (FW-1 and FW-2).*

Interface	Description	FW-1 address	FW-2 address
OPT1 (lagg0.110)	MGMT	10.10.10.220/24	10.10.10.221/24
OPT2 (lagg0.120)	DATABASE	10.10.20.220/24	10.10.20.221/24
OPT3 (lagg0.130)	APP_SERVER	10.10.30.220/24	10.10.30.221/24
OPT4 (lagg0.210)	DMZMANAGEMENT	10.20.10.220/24	10.20.10.221/24
OPT5 (lagg0.220)	DMZSERVICE	10.20.20.220/24	10.20.20.221/24
OPT6 (vtnet2)	PFSYNC	10.254.1.1/30	10.254.1.2/30

C.2.5 CARP virtual IP creation

Navigate to `Firewall` → `Virtual IPs` → `Add` and create five CARP virtual IPs on FW-1 only. FW-2 receives them automatically via XMLRPC synchronization. All VIPs use the same virtual IP password for CARP advertisement.

Table C.4: *CARP virtual gateway addresses (configured on FW-1).*

Interface	Type	Virtual address	VHID	Description
MGMT	CARP	10.10.10.254/24	10	Management gateway
DATABASE	CARP	10.10.20.254/24	20	Database gateway
APP_SERVER	CARP	10.10.30.254/24	30	App gateway
DMZMANAGEMENT	CARP	10.20.10.254/24	40	DMZ-Mgmt gateway
DMZSERVICE	CARP	10.20.20.254/24	50	DMZ-Service gateway

C.2.6 High availability synchronization

Navigate to `System` → `High Availability Sync` and configure state and configuration synchronization between the two firewalls over the dedicated PFSYNC interface.

On FW-1:

- Synchronize states: enabled
- Synchronize interface: PFSYNC
- pfsync peer IP: 10.254.1.2
- XMLRPC sync target IP: 10.254.1.2
- Items synchronized: Firewall Rules, Firewall Aliases, NAT Configuration, Static Routes, Virtual IPs

On FW-2:

- Synchronize states: enabled
- Synchronize interface: PFSYNC
- pfsync peer IP: 10.254.1.1
- XMLRPC section left empty (FW-2 is the sync target, not the source)

C.2.7 Firewall rules

Navigate to `Firewall` → `Rules` → `Floating` and apply the following rules in order:

1. **Block** – IPv4 ICMP – Source: DMZ zone networks → Destination: Trust zone networks (prevents DMZ from initiating ICMP into the Trust zone)
2. **Allow** – IPv4 ICMP – Source: any → Destination: any (permits all other ICMP traffic)
3. **Allow** – IPv4 TCP – Source: MANAGEMENT subnets → Destination: any (permits management plane SSH access)
4. **Allow** – IPv4 TCP – Source: APP_SERVER subnets → Destination: DATABASE subnets (permits application-to-database traffic)
5. **Allow** – IPv4 TCP – Source: APP_SERVER subnets → Destination: DMZSERVICE subnets (permits application servers to reach DMZ web tier)
6. **Allow** – IPv4 TCP – Source: DMZMANAGEMENT subnets → Destination: DMZSERVICE subnets (permits DMZ management to DMZ service traffic)
7. **Block** – IPv4 TCP – Source: any → Destination: any (default-deny rule for all remaining TCP traffic)

C.3 ToR downlinks to servers

The following Arista CLI scripts configure server-facing access ports and Port-Channels (Po2–Po5) on each ToR switch.

C.3.1 Cabinet A – ToR A-1

Listing C.9: *ToR A-1: downlinks to Servers 1 and 2.*

```
enable
configure terminal

interface Ethernet1
  description Server1_NIC_M1
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server1_NIC_S1
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server2_NIC_M1
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server2_NIC_S1
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server1_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
```

```
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server1_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server2_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server2_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.2 Cabinet A – ToR A-2

Listing C.10: *ToR A-2: downlinks to Servers 1 and 2 (backup NICs).*

```
enable
configure terminal

interface Ethernet1
  description Server1_NIC_M2
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server1_NIC_S2
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server2_NIC_M2
  no shutdown
  channel-group 4 mode on
```

```
interface Ethernet4
  description Server2_NIC_S2
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server1_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server1_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server2_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server2_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.3 Cabinet B – ToR B-1

Listing C.11: *ToR B-1: downlinks to Servers 3 and 4.*

```
enable
configure terminal

interface Ethernet1
  description Server3_NIC_M1
  no shutdown
```

```
  channel-group 2 mode on

interface Ethernet2
  description Server3_NIC_S1
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server4_NIC_M1
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server4_NIC_S1
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server3_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server3_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server4_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server4_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.4 Cabinet B – ToR B-2

Listing C.12: *ToR B-2: downlinks to Servers 3 and 4 (backup NICs).*

```
enable
configure terminal

interface Ethernet1
  description Server3_NIC_M2
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server3_NIC_S2
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server4_NIC_M2
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server4_NIC_S2
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server3_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server3_SERVICE
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server4_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server4_SERVICE
```

```
  switchport
  switchport mode access
  switchport access vlan 120
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.5 Cabinet C – ToR C-1

Listing C.13: *ToR C-1: downlinks to Servers 5 and 6.*

```
enable
configure terminal

interface Ethernet1
  description Server5_NIC_M1
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server5_NIC_S1
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server6_NIC_M1
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server6_NIC_S1
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server5_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server5_SERVICE
  switchport
  switchport mode access
  switchport access vlan 130
  mlag 3
  spanning-tree portfast
  no shutdown
```

```
interface Port-Channel4
  description Server6_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server6_SERVICE
  switchport
  switchport mode access
  switchport access vlan 130
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.6 Cabinet C – ToR C-2

Listing C.14: *ToR C-2: downlinks to Servers 5 and 6 (backup NICs).*

```
enable
configure terminal

interface Ethernet1
  description Server5_NIC_M2
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server5_NIC_S2
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server6_NIC_M2
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server6_NIC_S2
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server5_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 2
  spanning-tree portfast
```

```
  no shutdown

interface Port-Channel3
  description Server5_SERVICE
  switchport
  switchport mode access
  switchport access vlan 130
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server6_MGMT
  switchport
  switchport mode access
  switchport access vlan 110
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server6_SERVICE
  switchport
  switchport mode access
  switchport access vlan 130
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.7 Cabinet D – ToR D-1

Listing C.15: *ToR D-1: downlinks to Servers 7 and 8 (DMZ).*

```
enable
configure terminal

interface Ethernet1
  description Server7_NIC_M1
  no shutdown
  channel-group 2 mode on

interface Ethernet2
  description Server7_NIC_S1
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server8_NIC_M1
  no shutdown
  channel-group 4 mode on

interface Ethernet4
```

```
  description Server8_NIC_S1
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server7_MGMT
  switchport
  switchport mode access
  switchport access vlan 210
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server7_SERVICE
  switchport
  switchport mode access
  switchport access vlan 220
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server8_MGMT
  switchport
  switchport mode access
  switchport access vlan 210
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server8_SERVICE
  switchport
  switchport mode access
  switchport access vlan 220
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.3.8 Cabinet D – ToR D-2

Listing C.16: *ToR D-2: downlinks to Servers 7 and 8 (DMZ, backup NICs).*

```
enable
configure terminal

interface Ethernet1
  description Server7_NIC_M2
  no shutdown
  channel-group 2 mode on
```

```
interface Ethernet2
  description Server7_NIC_S2
  no shutdown
  channel-group 3 mode on

interface Ethernet3
  description Server8_NIC_M2
  no shutdown
  channel-group 4 mode on

interface Ethernet4
  description Server8_NIC_S2
  no shutdown
  channel-group 5 mode on

interface Port-Channel2
  description Server7_MGMT
  switchport
  switchport mode access
  switchport access vlan 210
  mlag 2
  spanning-tree portfast
  no shutdown

interface Port-Channel3
  description Server7_SERVICE
  switchport
  switchport mode access
  switchport access vlan 220
  mlag 3
  spanning-tree portfast
  no shutdown

interface Port-Channel4
  description Server8_MGMT
  switchport
  switchport mode access
  switchport access vlan 210
  mlag 4
  spanning-tree portfast
  no shutdown

interface Port-Channel5
  description Server8_SERVICE
  switchport
  switchport mode access
  switchport access vlan 220
  mlag 5
  spanning-tree portfast
  no shutdown

end
write memory
```

C.4 ToR initial setup

The following scripts configure hostname, VLANs, management SVI, SSH access, and MLAG peering on each ToR switch before server downlink ports are applied.

C.4.1 Management and VLAN setup

ToR A-1

Listing C.17: *ToR A-1: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORA-1

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.200/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR A-2

Listing C.18: *ToR A-2: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORA-2

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
```

```
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.201/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR B-1

Listing C.19: *ToR B-1: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORB-1

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.202/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR B-2

Listing C.20: *ToR B-2: initial management and VLAN setup.*

```
enable
configure terminal
```

```
hostname TORB-2

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.203/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR C-1

Listing C.21: *ToR C-1: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORC-1

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.204/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123
```

```
end
write memory
```

ToR C-2

Listing C.22: *ToR C-2: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORC-2

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.205/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR D-1

Listing C.23: *ToR D-1: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORD-1

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
```

```
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.206/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

ToR D-2

Listing C.24: *ToR D-2: initial management and VLAN setup.*

```
enable
configure terminal

hostname TORD-2

vlan 110
  name MANAGEMENT
vlan 120
  name DATABASE
vlan 130
  name APP
vlan 210
  name DMZ_MANAGEMENT
vlan 220
  name DMZ_SERVICE

interface Vlan110
  ip address 10.10.10.207/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

C.4.2 MLAG setup

ToR A-1

Listing C.25: *ToR A-1: MLAG peer-link and domain configuration.*

```
enable
configure terminal
```

```
interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.1/30
  no shutdown

mlag configuration
  domain-id MLAG-TORA
  local-interface Vlan110
  peer-address 10.10.10.201
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 1

end
write memory
```

ToR A-2

Listing C.26: *ToR A-2: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
```

```
  no switchport
  ip address 172.16.255.2/30
  no shutdown

mlag configuration
  domain-id MLAG-TORA
  local-interface Vlan110
  peer-address 10.10.10.200
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 100

end
write memory
```

ToR B-1

Listing C.27: *ToR B-1: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.5/30
  no shutdown

mlag configuration
  domain-id MLAG-TORB
  local-interface Vlan110
  peer-address 10.10.10.203
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 1

end
write memory
```

ToR B-2

Listing C.28: *ToR B-2: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.6/30
  no shutdown

mlag configuration
  domain-id MLAG-TORB
  local-interface Vlan110
  peer-address 10.10.10.202
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 100

end
write memory
```

ToR C-1

Listing C.29: *ToR C-1: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
```

```
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.9/30
  no shutdown

mlag configuration
  domain-id MLAG-TORC
  local-interface Vlan110
  peer-address 10.10.10.205
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 1

end
write memory
```

ToR C-2

Listing C.30: *ToR C-2: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.10/30
  no shutdown

mlag configuration
  domain-id MLAG-TORC
  local-interface Vlan110
  peer-address 10.10.10.204
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
```

```
  primary-priority 100

end
write memory
```

ToR D-1

Listing C.31: *ToR D-1: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk

interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.13/30
  no shutdown

mlag configuration
  domain-id MLAG-TORD
  local-interface Vlan110
  peer-address 10.10.10.207
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 1

end
write memory
```

ToR D-2

Listing C.32: *ToR D-2: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Port-Channel1
  switchport
  switchport mode trunk
```

```
interface Ethernet7
  description MLAG_PEER_1
  no shutdown
  channel-group 1 mode active

interface Ethernet8
  description MLAG_PEER_2
  no shutdown
  channel-group 1 mode active

interface Ethernet9
  description MLAG_DAD
  no switchport
  ip address 172.16.255.14/30
  no shutdown

mlag configuration
  domain-id MLAG-TORD
  local-interface Vlan110
  peer-address 10.10.10.206
  peer-link Port-Channel1
  dual-primary detection delay 5 action errdisable all-interfaces
  primary-priority 100

end
write memory
```

C.5 ToR uplinks to EoRs

The following scripts configure the MLAG uplink (Port-Channel6) from each ToR to both EoR switches. Cabinets A and B trunk VLANs 110 and 120; Cabinet C trunks 110 and 130; Cabinet D trunks 210 and 220.

C.5.1 ToR A-1

Listing C.33: *ToR A-1: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
```

```
  no shutdown

end
write memory
```

C.5.2 ToR A-2

Listing C.34: *ToR A-2: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
  no shutdown

end
write memory
```

C.5.3 ToR B-1

Listing C.35: *ToR B-1: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
```

```
  no shutdown

end
write memory
```

C.5.4 ToR B-2

Listing C.36: *ToR B-2: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
  no shutdown

end
write memory
```

C.5.5 ToR C-1

Listing C.37: *ToR C-1: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 6
```

```
  no shutdown

end
write memory
```

C.5.6 ToR C-2

Listing C.38: *ToR C-2: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 6
  no shutdown

end
write memory
```

C.5.7 ToR D-1

Listing C.39: *ToR D-1: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 210,220
  mlag 6
```

```
  no shutdown

end
write memory
```

C.5.8 ToR D-2

Listing C.40: *ToR D-2: uplinks to EoR-1 and EoR-2.*

```
enable
configure terminal

interface Ethernet5
  description Uplink_EoR1
  no shutdown
  channel-group 6 mode on

interface Ethernet6
  description Uplink_EoR2
  no shutdown
  channel-group 6 mode on

interface Port-Channel6
  description Uplink_EoR_MLAG
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 210,220
  mlag 6
  no shutdown

end
write memory
```

C.6 EoRs Initial Setup

The following scripts configure hostname, VLANs, management SVI, and SSH access on each EoR switch.

C.6.1 EoR 1

Listing C.41: *EoR 1: initial management and VLAN setup.*

```
enable
configure terminal

hostname EOR1

vlan 110
  name MANAGEMENT

interface Vlan110
  ip address 10.10.10.208/24
  no shut

management ssh
  no shutdown
```

```
username admin privilege 15 secret Admin@123

end
write memory
```

C.6.2 EoR 2

Listing C.42: *EoR 2: initial management and VLAN setup.*

```
enable
configure terminal

hostname EOR2

vlan 110
  name MANAGEMENT

interface Vlan110
  ip address 10.10.10.209/24
  no shutdown

management ssh
  no shutdown

username admin privilege 15 secret Admin@123

end
write memory
```

C.7 EoRs MLAG Setup

The following scripts configure the MLAG peer-link and domain configuration on each EoR switch.

C.7.1 EoR 1

Listing C.43: *EoR 1: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Ethernet23
  no switchport
  ip address 172.16.255.1/30
  no shutdown

interface Ethernet21
  no shutdown
  channel-group 1 mode active

interface Ethernet22
  no shutdown
  channel-group 1 mode active

interface Port-Channel1
  description MLAG-PEER-LINK
  switchport
```

```
  switchport mode trunk
  no shutdown

mlag configuration
  domain-id MLAG-EOR
  local-interface Vlan110
  peer-address 10.10.10.209
  peer-link Port-Channel1
  primary-priority 1
  dual-primary detection delay 5 action errdisable all-interfaces

end
write memory
```

C.7.2 EoR 2

Listing C.44: *EoR 2: MLAG peer-link and domain configuration.*

```
enable
configure terminal

interface Ethernet23
  no switchport
  ip address 172.16.255.2/30
  no shutdown

interface Ethernet21
  no shutdown
  channel-group 1 mode active

interface Ethernet22
  no shutdown
  channel-group 1 mode active

interface Port-Channel1
  description MLAG-PEER-LINK
  switchport
  switchport mode trunk
  no shutdown

mlag configuration
  domain-id MLAG-EOR
  local-interface Vlan110
  peer-address 10.10.10.208
  peer-link Port-Channel1
  primary-priority 100
  dual-primary detection delay 5 action errdisable all-interfaces

end
write memory
```

C.8 EoRs Downlinks to ToRs

The following scripts configure the downlink Port-Channels from each EoR to the ToR switches in each cabinet.

C.8.1 EoR 1

Listing C.45: *EoR 1: downlinks to ToR switches.*

```
enable
configure terminal

! Cabinet A
interface Ethernet1
  description Downlink_TorA1
  no shutdown
  channel-group 3 mode on

interface Ethernet2
  description Downlink_TorA2
  no shutdown
  channel-group 4 mode on

interface Port-Channel3
  description TorA1_MLAG3
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 3
  no shutdown

interface Port-Channel4
  description TorA2_MLAG4
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 4
  no shutdown

! Cabinet B
interface Ethernet3
  description Downlink_TorB1
  no shutdown
  channel-group 5 mode on

interface Ethernet4
  description Downlink_TorB2
  no shutdown
  channel-group 6 mode on

interface Port-Channel5
  description TorB1_MLAG5
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 5
  no shutdown

interface Port-Channel6
  description TorB2_MLAG6
  switchport
```

```
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
  no shutdown

! Cabinet C
interface Ethernet5
  description Downlink_TorC1
  no shutdown
  channel-group 7 mode on

interface Ethernet6
  description Downlink_TorC2
  no shutdown
  channel-group 8 mode on

interface Port-Channel7
  description TorC1_MLAG7
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 7
  no shutdown

interface Port-Channel8
  description TorC2_MLAG8
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 8
  no shutdown

! Cabinet D
interface Ethernet7
  description Downlink_TorD1
  no shutdown
  channel-group 9 mode on

interface Ethernet8
  description Downlink_TorD2
  no shutdown
  channel-group 10 mode on

interface Port-Channel9
  description TorD1_MLAG9
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 210,220
  mlag 9
  no shutdown

interface Port-Channel10
  description TorD2_MLAG10
  switchport
  switchport mode trunk
```

```
  switchport trunk allowed vlan 210,220
  mlag 10
  no shutdown

end
write memory
```

C.8.2 EoR 2

Listing C.46: *EoR 2: downlinks to ToR switches.*

```
enable
configure terminal

! Cabinet A
interface Ethernet1
  description Downlink_TorA1
  no shutdown
  channel-group 3 mode on

interface Ethernet2
  description Downlink_TorA2
  no shutdown
  channel-group 4 mode on

interface Port-Channel3
  description TorA1_MLAG3
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 3
  no shutdown

interface Port-Channel4
  description TorA2_MLAG4
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 4
  no shutdown

! Cabinet B
interface Ethernet3
  description Downlink_TorB1
  no shutdown
  channel-group 5 mode on

interface Ethernet4
  description Downlink_TorB2
  no shutdown
  channel-group 6 mode on

interface Port-Channel5
  description TorB1_MLAG5
  switchport
```

```
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 5
  no shutdown

interface Port-Channel6
  description TorB2_MLAG6
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,120
  mlag 6
  no shutdown

! Cabinet C
interface Ethernet5
  description Downlink_TorC1
  no shutdown
  channel-group 7 mode on

interface Ethernet6
  description Downlink_TorC2
  no shutdown
  channel-group 8 mode on

interface Port-Channel7
  description TorC1_MLAG7
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 7
  no shutdown

interface Port-Channel8
  description TorC2_MLAG8
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 110,130
  mlag 8
  no shutdown

! Cabinet D
interface Ethernet7
  description Downlink_TorD1
  no shutdown
  channel-group 9 mode on

interface Ethernet8
  description Downlink_TorD2
  no shutdown
  channel-group 10 mode on

interface Port-Channel9
  description TorD1_MLAG9
  switchport
  switchport mode trunk
```

```
  switchport trunk allowed vlan 210,220
  mlag 9
  no shutdown

interface Port-Channel10
  description TorD2_MLAG10
  switchport
  switchport mode trunk
  switchport trunk allowed vlan 210,220
  mlag 10
  no shutdown

end
write memory
```

APPENDIX D

APPENDIX D: CONNECTIVITY TEST RESULTS

D.1 Basic connectivity tests

```
root@kvm:~# ping 10.10.10.2
PING 10.10.10.2 (10.10.10.2) 56(84) bytes of data.
64 bytes from 10.10.10.2: icmp_seq=1 ttl=64 time=7.100 ms
64 bytes from 10.10.10.2: icmp_seq=2 ttl=64 time=1.50 ms
64 bytes from 10.10.10.2: icmp_seq=3 ttl=64 time=1.18 ms
64 bytes from 10.10.10.2: icmp_seq=4 ttl=64 time=1.26 ms
^C
--- 10.10.10.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 1.180/2.984/7.997/2.896 ms
root@kvm:~#
```

Figure D.1: *Ping test: Server 1 → Server 2 (management, 10.10.10.2).*

```
root@kvm:~# ping 10.10.20.2
PING 10.10.20.2 (10.10.20.2) 56(84) bytes of data.
64 bytes from 10.10.20.2: icmp_seq=1 ttl=64 time=3.27 ms
64 bytes from 10.10.20.2: icmp_seq=2 ttl=64 time=1.49 ms
64 bytes from 10.10.20.2: icmp_seq=3 ttl=64 time=1.35 ms
64 bytes from 10.10.20.2: icmp_seq=4 ttl=64 time=1.29 ms
64 bytes from 10.10.20.2: icmp_seq=5 ttl=64 time=1.53 ms
^C
--- 10.10.20.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 11ms
rtt min/avg/max/mdev = 1.291/1.784/3.267/0.748 ms
```

Figure D.2: *Ping test: Server 1 → Server 2 (service, 10.10.20.2).*

D.2 Intra-zone connectivity tests

```
root@kvm:~# ping 10.10.20.3
PING 10.10.20.3 (10.10.20.3) 56(84) bytes of data.
64 bytes from 10.10.20.3: icmp_seq=1 ttl=64 time=4.90 ms
64 bytes from 10.10.20.3: icmp_seq=2 ttl=64 time=5.04 ms
64 bytes from 10.10.20.3: icmp_seq=3 ttl=64 time=3.89 ms
^C
--- 10.10.20.3 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6ms
rtt min/avg/max/mdev = 3.888/4.610/5.039/0.516 ms
root@kvm:~#
```

Figure D.3: *Ping test: Server 1 → Server 3 (service, 10.10.20.3).*

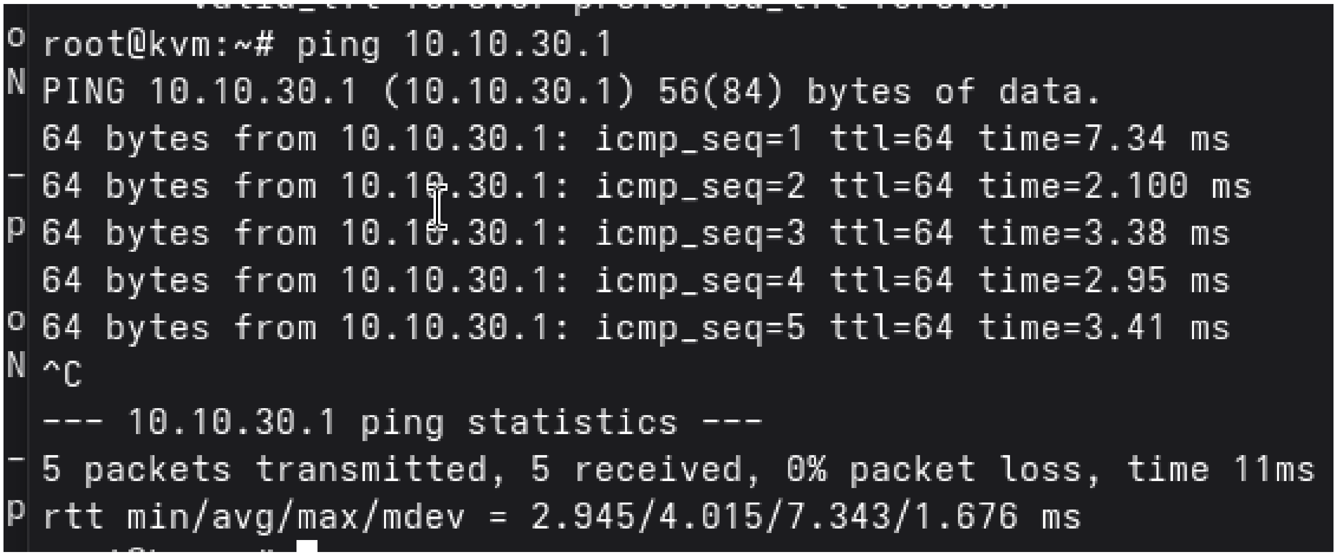


Figure D.4: *Ping test: Server 1 → Server 5 (service, 10.10.30.1).*

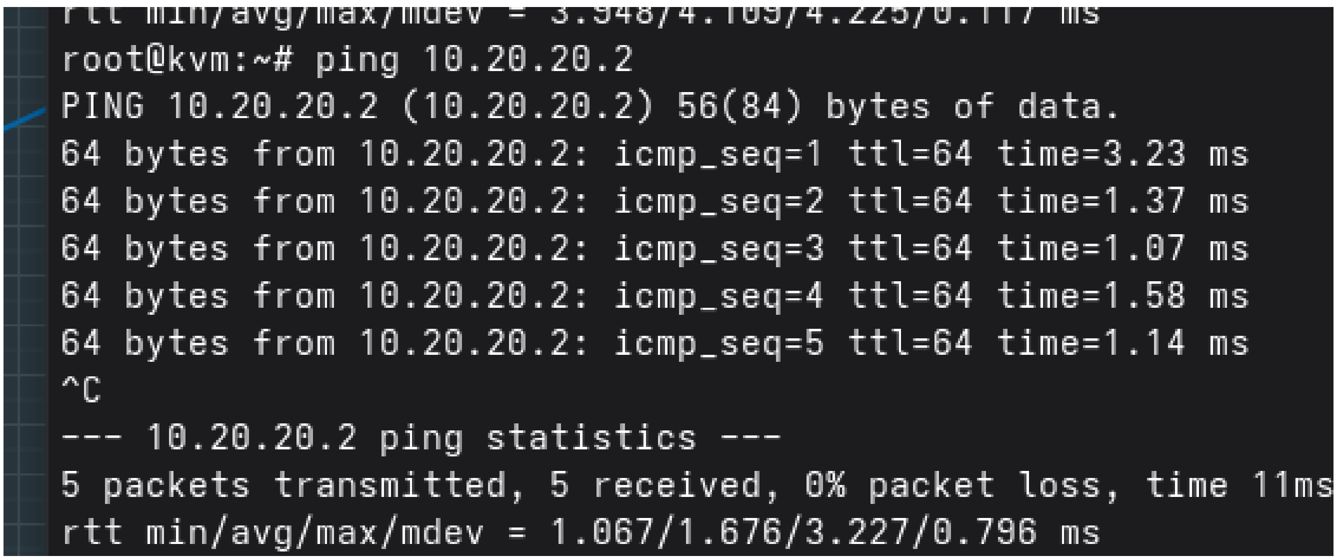


Figure D.5: *Ping test: Server 7 → Server 8 (service, 10.20.20.2).*

D.3 Inter-zone security tests

```
rtt min/avg/max/mdev = 2.945/4.015/7.343/1.676 ms
root@kvm:~# ping 10.20.10.1
PING 10.20.10.1 (10.20.10.1) 56(84) bytes of data.
64 bytes from 10.20.10.1: icmp_seq=1 ttl=63 time=4.44 ms
64 bytes from 10.20.10.1: icmp_seq=2 ttl=63 time=4.31 ms
64 bytes from 10.20.10.1: icmp_seq=3 ttl=63 time=4.10 ms
64 bytes from 10.20.10.1: icmp_seq=4 ttl=63 time=3.93 ms
^C
--- 10.20.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 3.928/4.194/4.441/0.201 ms
```

Figure D.6: *Ping test: Server 1 → Server 7 (management, 10.20.10.1), permitted.*

```
root@kvm:~# ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
^C
--- 10.10.10.1 ping statistics ---
7 packets transmitted, 0 received, 100% packet loss, time 149ms

root@kvm:~#
```

Figure D.7: *Ping test: Server 7 → Server 1 (management, 10.10.10.1), blocked.*

APPENDIX E

APPENDIX E: CONFIGURATION VERIFICATION SCREENSHOTS

E.1 EoR-1 (Arista vEOS)

```
EOR1#show vlan
VLAN  Name                             Status    Ports
----- -------------------------------- --------- -------------------------------
1     default                          active
110   MANAGEMENT                       active    Cpu, Po1, Po3, Po4, Po5, Po6
                                                 Po7, Po8, Po20, Po30
120   DATABASE                         active    Po1, Po3, Po4, Po5, Po6, Po20
                                                 Po30
130   APP                              active    Po1, Po7, Po8, Po20, Po30
210   DMZ_MANAGEMENT                   active    Po1, Po9, Po10, Po20, Po30
220   DMZ_SERVICE                      active    Po1, Po9, Po10, Po20, Po30
```

Figure E.1: *EoR-1: VLANs (`show vlan`).*

```
EOR1#show ip interface brief
                                                                          Address
Interface        IP Address          Status      Protocol         MTU    Owner
---------------- ------------------- ----------- ------------- --------- -------
Ethernet23       172.16.255.1/30     up          up              1500
Management1      unassigned          down        down            1500
Vlan110          10.10.10.208/24     up          up              1500
```

Figure E.2: *EoR-1: IP addressing (`show ip interface brief`).*

```
EOR1#show interfaces status
Port       Name           Status       Vlan     Duplex Speed  Type
Et1        Downlink_TorA1 connected    in Po3   full   1G     EbraTestPhyPort
Et2        Downlink_TorA2 connected    in Po4   full   1G     EbraTestPhyPort
Et3        Downlink_TorB1 connected    in Po5   full   1G     EbraTestPhyPort
Et4        Downlink_TorB2 connected    in Po6   full   1G     EbraTestPhyPort
Et5        Downlink_TorC1 connected    in Po7   full   1G     EbraTestPhyPort
Et6        Downlink_TorC2 connected    in Po8   full   1G     EbraTestPhyPort
Et7        Downlink_TorD1 connected    in Po9   full   1G     EbraTestPhyPort
Et8        Downlink_TorD2 connected    in Po10  full   1G     EbraTestPhyPort
Et9                       notconnect   1        full   1G     EbraTestPhyPort
Et10                      notconnect   1        full   1G     EbraTestPhyPort
Et11                      notconnect   1        full   1G     EbraTestPhyPort
Et12                      notconnect   1        full   1G     EbraTestPhyPort
Et13                      notconnect   1        full   1G     EbraTestPhyPort
Et14                      notconnect   1        full   1G     EbraTestPhyPort
Et15                      notconnect   1        full   1G     EbraTestPhyPort
Et16                      notconnect   1        full   1G     EbraTestPhyPort
Et17                      notconnect   1        full   1G     EbraTestPhyPort
Et18                      notconnect   1        full   1G     EbraTestPhyPort
Et19                      notconnect   1        full   1G     EbraTestPhyPort
Et20                      notconnect   1        full   1G     EbraTestPhyPort
Et21                      connected    in Po1   full   1G     EbraTestPhyPort
Et22                      connected    in Po1   full   1G     EbraTestPhyPort
Et23                      connected    routed   full   1G     EbraTestPhyPort
Et24       to-FW1-G1/0/0  connected    in Po20  full   1G     EbraTestPhyPort
Et25       to-FW2-G1/0/0  connected    in Po30  full   1G     EbraTestPhyPort
Et26                      notconnect   1        full   1G     EbraTestPhyPort
Et27                      notconnect   1        full   1G     EbraTestPhyPort
Et28                      notconnect   1        full   1G     EbraTestPhyPort
Et29                      notconnect   1        full   1G     EbraTestPhyPort
Ma1                       notconnect   routed   a-full a-1G   10/100/1000
Po1        MLAG-PEER-LINK connected    trunk    full   2G     N/A
Po3        TorA1_MLAG3    connected    trunk    full   2G     N/A
Po4        TorA2_MLAG4    connected    trunk    full   2G     N/A
Po5        TorB1_MLAG5    connected    trunk    full   2G     N/A
Po6        TorB2_MLAG6    connected    trunk    full   2G     N/A
Po7        TorC1_MLAG7    connected    trunk    full   2G     N/A
Po8        TorC2_MLAG8    connected    trunk    full   2G     N/A
Po9        TorD1_MLAG9    connected    trunk    full   2G     N/A
Po10       TorD2_MLAG10   connected    trunk    full   2G     N/A
Po20       MLAG-to-FW1    connected    trunk    full   2G     N/A
Po30       MLAG-to-FW2    connected    trunk    full   2G     N/A
```

Figure E.3: *EoR-1: Interfaces and Port-Channels.*

```
EOR1#show mlag
MLAG Configuration:
domain-id                          :          MLAG-EOR
local-interface                    :           Vlan110
peer-address                       :      10.10.10.209
peer-link                          :      Port-Channel1
hb-peer-address                    :           0.0.0.0
peer-config                        :        consistent

MLAG Status:
state                              :            Active
negotiation status                 :         Connected
peer-link status                   :                Up
local-int status                   :                Up
system-id                          :   52:17:5d:db:36:98
dual-primary detection             :          Disabled
dual-primary interface errdisabled :             False

MLAG Ports:
Disabled                           :                 0
Configured                         :                 0
Inactive                           :                 0
Active-partial                     :                 0
Active-full                        :                10
```

Figure E.4: *EoR-1: MLAG (`show mlag`).*

E.2 ToR A-1 (Arista vEOS)

```
TORA-1#show vlan brief
VLAN  Name                             Status    Ports
----- -------------------------------- --------- -------------------------------
1     default                          active    Et10, Et11, Et12, Et13, Et14
                                                 Et15, Et16, Et17, Et18, Et19
                                                 Po1
110   MANAGEMENT                       active    Cpu, Po1, Po2, Po4, Po6
120   DATABASE                         active    Po1, Po3, Po5, Po6
130   APP                              active    Po1
210   DMZ_MANAGEMENT                   active    Po1
220   DMZ_SERVICE                      active    Po1
```

Figure E.5: *ToR A-1: VLANs (`show vlan brief`).*

```
TORA-1#show ip interface  br
                                                                            Address
Interface       IP Address          Status      Protocol         MTU    Owner
--------------- ------------------- ----------- ------------- --------- -------
Ethernet9       172.16.255.1/30     up          up              1500
Management1     unassigned          down        down            1500
Vlan110         10.10.10.200/24     up          up              1500
```

Figure E.6: *ToR A-1: IP addressing.*

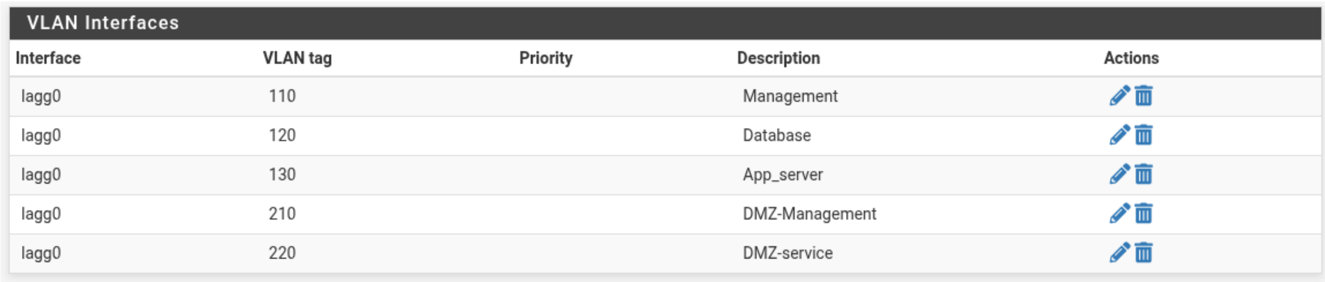
```
TORA-1# show interfaces status
Port       Name            Status       Vlan     Duplex Speed  Type
Et1        Server1_NIC_M1  connected    in Po2   full   1G     EbraTestPhyPort
Et2        Server1_NIC_S1  connected    in Po3   full   1G     EbraTestPhyPort
Et3        Server2_NIC_M1  connected    in Po4   full   1G     EbraTestPhyPort
Et4        Server2_NIC_S1  connected    in Po5   full   1G     EbraTestPhyPort
Et5        Uplink_EoR1     connected    in Po6   full   1G     EbraTestPhyPort
Et6        Uplink_EoR2     connected    in Po6   full   1G     EbraTestPhyPort
Et7        MLAG_PEER_1     connected    in Po1   full   1G     EbraTestPhyPort
Et8        MLAG_PEER_2     connected    in Po1   full   1G     EbraTestPhyPort
Et9        MLAG_DAD        connected    routed   full   1G     EbraTestPhyPort
Et10                       connected    1        full   1G     EbraTestPhyPort
Et11                       connected    1        full   1G     EbraTestPhyPort
Et12                       connected    1        full   1G     EbraTestPhyPort
Et13                       connected    1        full   1G     EbraTestPhyPort
Et14                       connected    1        full   1G     EbraTestPhyPort
Et15                       connected    1        full   1G     EbraTestPhyPort
Et16                       connected    1        full   1G     EbraTestPhyPort
Et17                       connected    1        full   1G     EbraTestPhyPort
Et18                       connected    1        full   1G     EbraTestPhyPort
Et19                       connected    1        full   1G     EbraTestPhyPort
Ma1                        notconnect   routed   a-full a-1G   10/100/1000
Po1                        connected    trunk    full   2G     N/A
Po2        Server1_MGMT    connected    110      full   2G     N/A
Po3        Server1_SERVICE connected    120      full   2G     N/A
Po4        Server2_MGMT    connected    110      full   2G     N/A
Po5        Server2_SERVICE connected    120      full   2G     N/A
Po6        Uplink_EoR_MLAG connected    trunk    full   4G     N/A
```

Figure E.7: *ToR A-1: Interfaces and Port-Channels.*

```
TORA-1#show mlag
MLAG Configuration:
domain-id                          :           MLAG-TORA
local-interface                    :             Vlan110
peer-address                       :        10.10.10.201
peer-link                          :        Port-Channel1
hb-peer-address                    :             0.0.0.0
peer-config                        :          consistent

MLAG Status:
state                              :              Active
negotiation status                 :           Connected
peer-link status                   :                  Up
local-int status                   :                  Up
system-id                          :   52:9f:42:cd:61:c2
dual-primary detection             :            Disabled
dual-primary interface errdisabled :               False

MLAG Ports:
Disabled                           :                   0
Configured                         :                   0
Inactive                           :                   0
Active-partial                     :                   0
Active-full                        :                   5

TORA-1#
```

Figure E.8: *ToR A-1: MLAG (`show mlag`).*

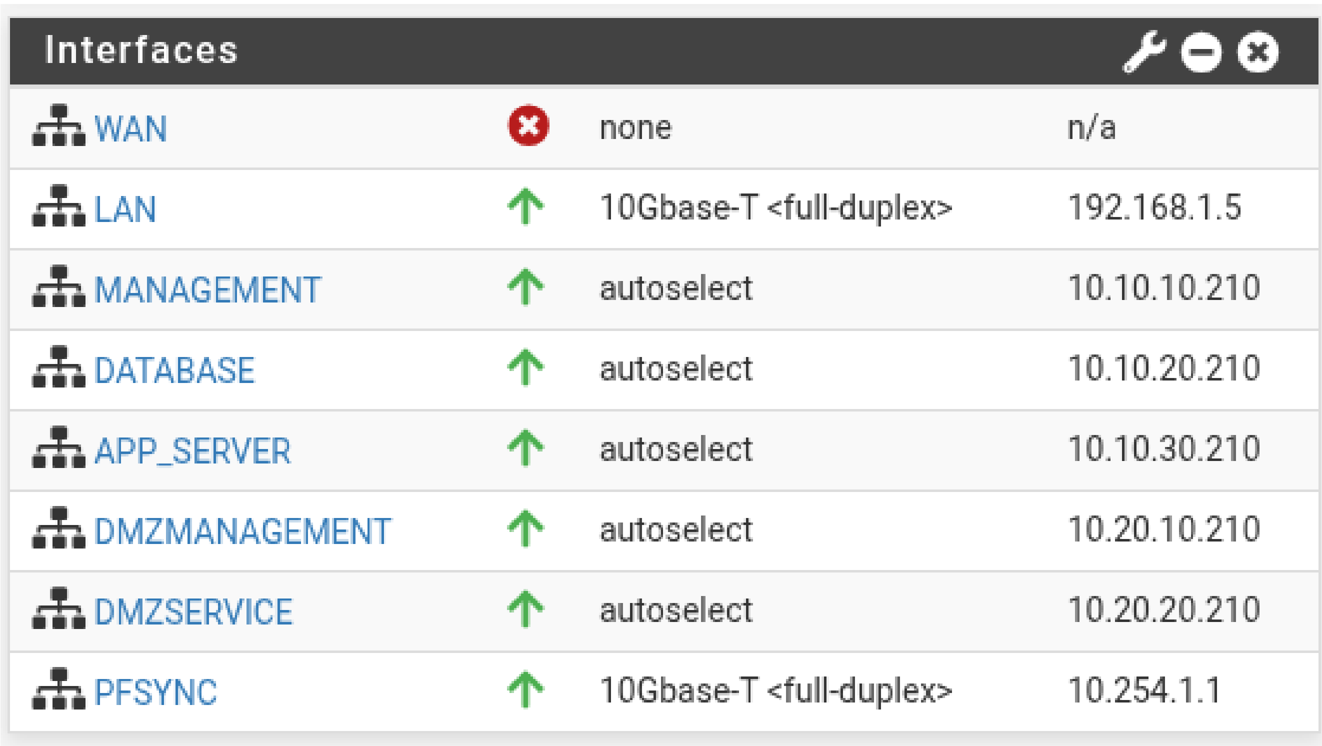
E.3 Firewalls (pfSense)



VLAN Interfaces

Interface	VLAN tag	Priority	Description	Actions
lagg0	110		Management	
lagg0	120		Database	
lagg0	130		App_server	
lagg0	210		DMZ-Management	
lagg0	220		DMZ-service	

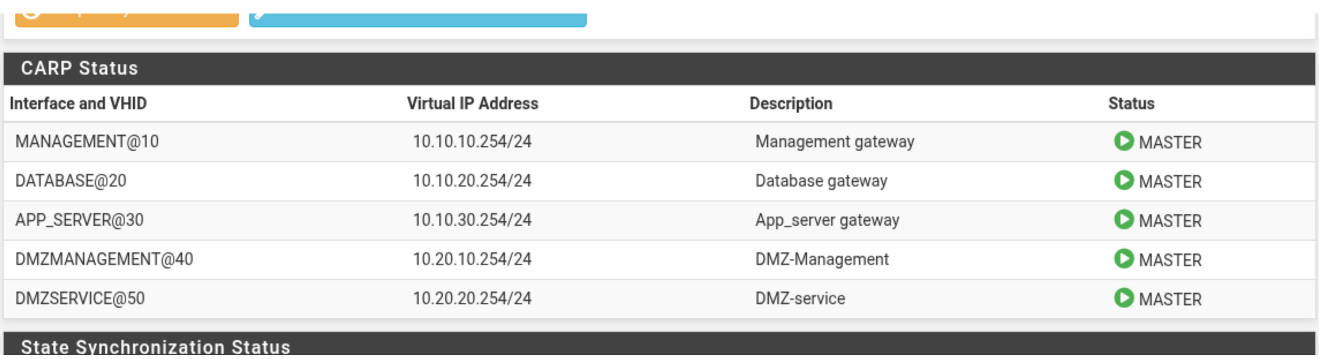
Figure E.9: *FW-1: VLAN interfaces.*



Interfaces

WAN		none	n/a
LAN		10Gbase-T <full-duplex>	192.168.1.5
MANAGEMENT		autoselect	10.10.10.210
DATABASE		autoselect	10.10.20.210
APP_SERVER		autoselect	10.10.30.210
DMZMANAGEMENT		autoselect	10.20.10.210
DMZSERVICE		autoselect	10.20.20.210
PFSYNC		10Gbase-T <full-duplex>	10.254.1.1

Figure E.10: *FW-1: Interfaces and IPs.*



CARP Status

Interface and VHID	Virtual IP Address	Description	Status
MANAGEMENT@10	10.10.10.254/24	Management gateway	MASTER
DATABASE@20	10.10.20.254/24	Database gateway	MASTER
APP_SERVER@30	10.10.30.254/24	App_server gateway	MASTER
DMZMANAGEMENT@40	10.20.10.254/24	DMZ-Management	MASTER
DMZSERVICE@50	10.20.20.254/24	DMZ-service	MASTER

State Synchronization Status

Figure E.11: *FW-1: CARP (master).*

CARP Status

Interface and VHID	Virtual IP Address	Description	Status
MANAGEMENT@10	10.10.10.254/24	Management gateway	BACKUP
DATABASE@20	10.10.20.254/24	Database gateway	BACKUP
APP_SERVER@30	10.10.30.254/24	App_server gateway	BACKUP
DMZMANAGEMENT@40	10.20.10.254/24	DMZ-Management	BACKUP
DMZSERVICE@50	10.20.20.254/24	DMZ-service	BACKUP

State Synchronization Status

Figure E.12: *FW-2: CARP (backup).*

Rules (Drag to Change Order)

States	Interfaces	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
0/0 B	Any	IPv4 ICMP any	DMZ_zone networks	*	Trust_zone networks	*	*	none			
0/0 B	Any	IPv4 ICMP any	*	*	*	*	*	none			
0/0 B	Any	IPv4 TCP	MANAGEMENT subnets	*	*	*	*	none			
0/0 B	Any	IPv4 TCP	APP_SERVER subnets	*	DATABASE subnets	*	*	none			
0/0 B	Any	IPv4 TCP	APP_SERVER subnets	*	DMZSERVICE subnets	*	*	none			
0/0 B	Any	IPv4 TCP	DMZMANAGEMENT subnets	*	DMZSERVICE subnets	*	*	none			
0/900 B	Any	IPv4 TCP	*	*	*	*	*	none			

Add Add Delete Toggle Save Separator

Figure E.13: *FW-1: Firewall rules.*

E.4 Server 1 (Linux)

```
root@kvm:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast master bond0 state UP group default qlen 1000
    link/ether 50:92:49:00:0b:00 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast master bond1 state UP group default qlen 1000
    link/ether 50:92:49:00:0b:01 brd ff:ff:ff:ff:ff:ff
4: eth2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast master bond0 state UP group default qlen 1000
    link/ether 50:92:49:00:0b:00 brd ff:ff:ff:ff:ff:ff
5: eth3: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast master bond1 state UP group default qlen 1000
    link/ether 50:92:49:00:0b:01 brd ff:ff:ff:ff:ff:ff
6: eth4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 50:92:49:00:0b:04 brd ff:ff:ff:ff:ff:ff
7: bond0: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 50:92:49:00:0b:00 brd ff:ff:ff:ff:ff:ff
    inet 10.10.10.1/24 brd 10.10.10.255 scope global bond0
       valid_lft forever preferred_lft forever
    inet6 fe80::5292:49ff:fe00:b00/64 scope link
       valid_lft forever preferred_lft forever
8: bond1: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 50:92:49:00:0b:01 brd ff:ff:ff:ff:ff:ff
    inet 10.10.20.1/24 brd 10.10.20.255 scope global bond1
       valid_lft forever preferred_lft forever
    inet6 fe80::5292:49ff:fe00:b01/64 scope link
       valid_lft forever preferred_lft forever
root@kvm:~#
```

Figure E.14: *Server 1: Bonding and IP assignments (`ip a`).*

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ABSTRACT

This dissertation, prepared for the Licence degree in ISIL at USTHB, presents the design and simulation of a modern data center network architecture suited to highly available enterprise workloads. The project adopts a spine-leaf topology with full-mesh connectivity between eight Top-of-Rack leaf switches and two End-of-Rack spine switches, complemented by zone-based security at the network edge. The infrastructure is realized in a virtual laboratory: PNETLab emulates the switching and firewall fabric, while VMware Workstation hosts Linux servers distributed across four cabinets.

The proposed facility organizes four cabinets into distinct roles and security zones. Cabinets A, B, and C form the Trust zone, hosting database and application servers; Cabinet D hosts web-facing services in the DMZ. Two perimeter firewalls operate in an active-standby cluster using CARP, enforcing strict separation so that external or DMZ compromise cannot reach internal databases without explicit policy. Each of the eight servers connects through dual management and service network interfaces, bonded for resilience and mapped to dedicated VLANs for management (110, 210), database (120), application (130), and web (220) traffic. At every layer (firewall, spine, and leaf), redundancy eliminates single points of failure: MLAG pairs at ToR and EoR and VRRP/CARP gateways.

The design is specified through functional and non-functional requirements, actor responsibilities, UML use-case and sequence models, a high-level topology, and low-level documentation including device inventory, IP addressing, and port-level cabling matrices. The implementation stack combines PNETLab, VMware Workstation Pro, Arista vEOS switches, pfSense firewalls, and Debian servers. Switches were configured with VLANs, trunk and access Port-Channels, MLAG domains, and management SVIs; firewalls with zone-based rules and default-deny policies; servers with active-backup bonding on management and service planes.

End-to-end validation combined connectivity testing and deliberate failure injection. ICMP tests confirmed allowed flows within and across cabinets on both management and service networks, as well as blocked DMZ-to-Trust management paths where policy required denial. Shutting down active firewalls, spine switches, and leaf switches demonstrated sub-second failover to standby nodes without loss of forwarding. Together, these results confirm that the architecture meets its redundancy, security, and manageability objectives.

The outcome is a coherent, fully documented, and rigorously tested blueprint, from topology and addressing through configuration templates and test evidence, that translates modern spine-leaf and zero-trust zoning principles into a production-ready enterprise data center design.